SMOKEFIN: AN IoT-ENABLED FISH SMOKING SYSTEM FOR SUSTAINABLE TINAPA PRODUCTION

A CpE Practice and Design Project
Presented to the Faculty of the
College of Computer Studies
MINDORO STATE UNIVERSITY
Bongabong Campus
Labasan, Bongabong Oriental Mindoro

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Computer Engineering

by

DENMARK G. ACOSTA
JAYPERSON M. CAYA
YASME O. FAJARITO
JOHN AERON P. FRUELDA
RUZZEL A. LIVELO

May 2026

APPROVAL SHEET

This Design Project entitled **"SMOKEFIN: AN IOT-ENABLED FISH SMOKING SYSTEM FOR SUSTAINABLE TINAPA PRODUCTION"** prepared and submitted by **DENMARK G. ACOSTA, JAYPERSON M. CAYA, YASME O. FAJARITO, JOHN AERON P. FRUELDA,** and **RUZZEL A. LIVELO,** has been successfully presented and recommended for approval by the Advisory Committee in partial fulfillment of the requirements for the degree, **Bachelor of Science in Computer Engineering.**

DR. EVELYN A. RODRIGUEZ, *DIT*
Adviser

PANELS OF EXAMINERS

ENGR. ARNOLD F. REANO
Major Critic

____
DATE

ENGR. FLOYD D. DE VELA, *MIT*
Technical Critic

____
DATE

ENGR. ALONA L. LABAGUIS, *MSCpE*
Chairperson

____
DATE

Approved and accepted in partial fulfillment of the requirements for the degree, **Bachelor of Science in Computer Engineering.**

JOHN EDGAR S. ANTHONY, *DIT*
Dean, College of Computer Studies

____
DATE

ACKNOWLEDGEMENT

The successful completion of this design project, SMOKEFIN, would not have been possible without the invaluable support, guidance, and encouragement of many individuals who contributed throughout this journey.

Above all, we offer our deepest gratitude to God for granting us the strength, wisdom, and perseverance needed to overcome challenges and accomplish this project.

We would like to express our heartfelt appreciation to our parents, Mr. & Mrs. Acosta, Mr. & Mrs. Caya, Mr. & Mrs. Fajarito, and Mr. & Mrs. Fruelda, and Mr. and Mrs. Livelo, for their unwavering love, support, and encouragement, which inspired us to continue striving for success.

Our sincere thanks go to our adviser, Dr. Evelyn A. Rodriguez, for her patience, guidance, and dedication. Her mentorship played a significant role in the successful completion of this project.

We are also grateful to the panel members who evaluated our work. We extend special recognition to Engr. Floyd D. De Vela, Engr. Alona Labaguis, Engr. Arnold F. Reano, for their insightful comments, and valuable technical recommendations, and our College Dean, Mr. John Edgar Anthony, for his continuous support and leadership. Their contributions greatly enhanced the quality of our study.

We would also like to acknowledge the MinSU community for upholding the values of Responsibility, Involvement, Commitment, and Excellence, which have guided us throughout our academic journey.

Lastly, we thank our fellow classmates for their camaraderie, encouragement, and shared experiences that made this journey more meaningful.

To everyone who contributed, directly or indirectly, we extend our sincere and heartfelt gratitude.

D.G.A

J.M.C

Y.O.F

J.A.P.F

R.A.L

ABSTRACT

The SmokeFin project is a technological modernization of the traditional Filipino fish smoking process (Tinapa), developed to address the limitations of labor-intensive, inconsistent, and health-risking manual methods. This IoT-enabled fish smoking system integrates sensors for temperature, humidity, and smoke level monitoring, combined with a microcontroller-based architecture utilizing an Arduino Mega 2560 and ESP32 module for real-time data processing and wireless communication. The system enhances the smoking process through controlled heating elements, actuators, and a user-friendly mobile application for remote monitoring.

Performance evaluation was conducted in accordance with ISO 14159:2002 for hygienic machine design and ISO 25010:2023 for software quality, assessing criteria such as functionality, usability, efficiency, safety, and reliability. Results showed that SmokeFin reduces smoking duration from 3-5 hours to 1–3 hours while increasing production output from 5–7 kg to 8–14 kg per batch. The system received an overall evaluation rating of "Very Acceptable," highlighting its effectiveness in improving operational safety, consistency, and user experience. Ultimately, SmokeFin provides a safer, more efficient, and sustainable solution for small-scale fish processors and vendors, supporting food security, economic growth, and technological advancement while preserving the cultural practice of traditional Tinapa production.

Keywords: IoT, Fish Smoking System, Tinapa, SmokeFin, Automation, Real-Time Monitoring, Food Processing Technology, Sustainable Production.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF APPENDICES

Chapter I

PROJECT DEFINITION AND DESCRIPTION

Introduction

Fish plays an important role in the lives of many people living in coastal and riverside communities. It is not only a daily source of food but also a major source of livelihood for small-scale fishers and their families. However, fresh fish spoils quickly, especially in areas without access to refrigeration. To address this, Filipinos have traditionally turned to smoking fish, producing what is commonly known as Tinapa. This method helps extend the shelf life of fish while adding flavor and value, allowing communities to store, sell, and transport their catch more effectively.

Traditionally, fish smoking is done manually over open fires. This process takes a lot of time, requires constant monitoring, and exposes the fish processors often women and elders to smoke, heat, and other health hazards. While this method has been passed down for generations, it can be inconsistent and labor intensive. To improve this practice, a SmokeFin machine was developed. This machine is designed to make the smoking process safer, faster, and more consistent, helping small fish vendors produce higher quality products while reducing health risks and physical strain [1].

The development of an Internet of Things (IoT) based Tinapa machine aims to enhance fish smoking by automating the process and providing real-time data on temperature, humidity, and smoking duration. This system combines sensors with wireless technology to help improve quality, reduce waste, and ensure safer food production [2]. The researchers chose Adame Jr.'s Tinapahan located at Barangay San Isidro, Bongabong Oriental Mindoro as beneficiary which is one of the suppliers of Tinapa product in the town. Adame Jr.'s Tinapahan currently using traditional way of making Tinapa which requires manual process of smoking. Researchers believed that the project would bring benefit to their business and help them keep up with the advancement of technology. This innovation is also helpful for community cooperatives and small businesses who rely on Tinapa for income.

In addition, this project supports sustainable development goals such as SDG 2 known as Zero hunger, as the machine helps reduce food waste and extend fish shelf life, ensuring a more stable and reliable food supply for fish-based communities. It also supports SDG 8 known as Decent Work and Economic Growth, by improving efficiency, and safety in tinapa production, the system increases productivity and opportunities for local tinapa vendors. Moreover, the project also supports SDG 9 known as Industry, Innovation, and

Infrastructure by integrating IoT technology into traditional fish process, promoting innovation, modernizing local industries, and strengthening sustainable production infrastructure. The IoT-enabled Tinapa maker not only preserves fish but also upholds culture, supports local livelihoods, and builds a more sustainable future for fish-based communities.

Lastly, this project aligns with ISO standards to ensure safety, quality, and sustainability. ISO 14159:2002 known as safety of the Machinery – Hygiene Requirements for the design was integrated to ensure hygienic conditions were the machine consistently maintained during operation. Furthermore, this standard guided the design of pre-processing, processing, and post-processing stages, emphasizing cleanability and contamination prevention. By adhering to ISO principles, the design minimized hygiene risks by enabling effective removal of residues and microorganisms, thereby ensuring product safety and quality [3].

To further support the software quality assurance of the project's mobile application, ISO 2501025010:2011 known as Systems and Software Quality Requirements and Evaluation standard was integrated. This provides a comprehensive framework for identifying software strengths and weaknesses, allowing developers to systematically assess application performance based on both functional and non-functional requirements. By integrating this standard into the evaluation process, the developers were able to ensure that the application met user expectations while maintaining high standards of software reliability and operational efficiency [4].

Statement of the Requirements

This project aims to develop SmokeFin: An IoT Enabled Fish Smoking System for Sustainable Tinapa Production. Specifically, this project aimed to fulfill the following specific objectives:

1. To design an IoT-based fish smoking system that can monitor environmental conditions such as temperature, humidity, and smoke level to achieve uniform smoking results;
2. To develop an interface that enables users to remotely monitor the fish smoking process through a mobile application.
3. To integrate electronic components such as sensors, heating elements, actuators, and wireless communication modules to support automated monitoring and controlled operation of the fish smoking process, while providing

real-time system feedback through the machine's interface and mobile application; and

4. To test and evaluate the system's overall performance in accordance with the ISO 14159:2002 and ISO 25010:2023 standards.

Scope and Limitations

This study focused on designing and developing an IoT-enabled Tinapa maker that aim to improve and modernize the traditional fish smoking process. The system is designed primarily for small-scale tinapa production and integrates an ESP32 microcontroller to monitor and manage the smoking operation. The machine allows users to monitor the temperature, humidity, and smoke level through both an LCD interface installed on the machine and a mobile application.

The LCD interface serves as the main control unit of the system, where users can directly select the smoking time based on preset time options recommended by the system. The LCD also displays real-time data on temperature, humidity, smoke level, and cooking duration to ensure proper monitoring during operation. Meanwhile, the mobile application is limited to monitoring functions only, allowing users to view temperature, humidity, smoke level, cooking time, and cooking history. The only control feature available in the mobile application is the activation of the water sprinkler to wet the wood dust in the smoke box so that it does not continue to rot, that it can be dried and use again next time. Water sprinkler is only activated when the process of smoking is already done and there will be no more fish to be smoke in that particular day.

The machine is designed to perform only the smoking process. Manual loading of wood dust is required, and other pre-processing steps such as brining, cleaning, and size sorting of fish are not included in the system's operation. The machine contains seven (7) fish racks, with each rack capable of holding approximately 20 to 25 pieces of fish depending on their size, allowing an estimated total capacity of 270 to 315 fish per smoking batch. The smoking duration typically ranges from one (1) to one and a half (1.5) hours, depending on the desired level of doneness or dryness of the tinapa.

Despite its functional features, the system has certain limitations. The project is intended only for small-scale use and is not designed to meet the requirements of large-scale or commercial fish smoking operations. The system does not automatically control or regulate the smoke level and humidity; these parameters are only monitored and displayed. In addition, the system relies on a stable internet connection for both the mobile application and the LCD interface to function properly. Monitoring and control of the machine through the mobile application, as well as real-time data display on the LCD, are only possible when the device is connected to the internet. Moreover, the machine does not include a feature that determines whether the fish is fully smoked or ready for consumption. The sign that the fish is fully smoked or ready is based on its color, it must be golden brown. Human supervision is still required to assess the final quality of the smoked fish and to ensure proper operation throughout the smoking process.

Significance of the Study

In this project, the SmokeFin modernized and improved the traditional process of making Tinapa. This innovation provides small-scale fish processors and vendors a safer, more consistent, and efficient way to preserve and add value to their catch. The system will allow users to monitor smoking conditions in real-time, reduce health risks, and potentially increase income through higher quality products. The following would benefit from the researchers' project:

Adame Jr.'s Tinapahan: This project will benefit the business of Mr. Danilo C. Adame Jr who currently uses the traditional method of making Tinapa that requires manual control of temperature, and time. With the SmokeFin Machine, these processes become automated and it also improves safety by reducing exposure to smoke and heat.

Public Small-Scale Fisherfolk and Vendors: This project will benefit local fishers and small businesses by giving them access to an easy-to-use system that increases their productivity, and adds value to their products through better quality control.

Coastal and Riverside Communities: These communities depend on fish not just for food but also for income. SmokFin will support them by improving food preservation practices, thereby contributing to food security and economic stability, especially in areas without access to reliable refrigeration.

Future Researchers: This study can serve as a useful reference for those aiming to develop similar systems or enhancements. It will open opportunities for expanding the scope to largescale operations, incorporating solar power, or improving food safety mechanisms.

Definition of Terms

Automation refers to the programmed control of the fish smoking process to reduce manual labor and ensure consistency in production.

Fish Smoking is the process of preserving fish by applying smoke and heat, modernized in this study through an IoT-based system.

Food Security ensures communities have access to safe, high-quality smoked fish with longer shelf life and reduced spoilage.

Internet of Things (IoT) is a network of connected devices that exchange data. In this study, it allows real-time monitoring of smoking conditions.

Prototype is a working model of the SmokeFin system developed and tested to evaluate its performance and effectiveness.

Real-Time Monitoring refers to the continuous tracking of temperature, humidity, and smoking duration during the fish smoking process.

Small-Scale Vendors refers to the individuals or groups who catch and sell fish in limited quantities; the main beneficiaries of the SmokeFin system.

SmokeFin refers to the IoT-enabled fish smoking system developed in this study to automate and improve Tinapa production.

Sustainability refers to balancing productivity, environmental safety, and livelihood support, which this project promotes through improved fish smoking practices.

Tinapa refers to the smoked fish product produced by curing fish and smoking it in order to prolong its shelf life.

Chapter II

TECHNICAL SURVEY REVIEW

The SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production highlights the advancement of traditional fish preservation methods through modern technology. This technical survey review emphasizes the importance of assessing existing practices in fish smoking and lot-based automation systems to guide the development of a sustainable and efficient solution for Tinapa production. By examining prior studies, tools, and standards, this research builds on proven methods while integrating sensors. microcontrollers, and wireless connectivity to optimize consistency, safety, and productivity. The review underscores the need to transform labor-intensive, health-risking manual processes into smart, automated systems that meet food safety, quality, and sustainability standards, providing a foundation for innovation tailored to the needs of small- scale fish vendors and coastal communities.

Existing Technology, Methods, and Patents

1. Traditional Fish Smoking Technology

Traditional fish smoking has been widely practiced in coastal and riverside communities. It involves the use of open fire or makeshift smoking chambers where fish are placed on bamboo racks and exposed to heat and smoke. While effective in extending shelf life, this method is labor-intensive, time-consuming, and exposes workers to health hazards such as smoke inhalation and burns. Moreover, inconsistent temperature and humidity often lead to varying product quality.

1.1 Chorkor Oven

The Chorkor oven is an improved fish smoking technology that uses local materials and stackable trays for higher capacity. It is fuel-efficient, using up to 80% less fuel compared to traditional ovens. This technology produces better-quality smoked fish, reduces health risks, and increases income opportunities for fish processors.

1.2 Cylindrical Metal Oven

The cylindrical metal oven is made from joined 44gallon steel drums and is widely used across Africa. It is light, portable, and provides support for multiple layers of fish. However, it is prone to rust, inefficient in heat management, and uncomfortable for processors due to high external heat.

1.3 Rectangular Mud Oven

The rectangular mud oven is built from mud with iron bars and wire mesh to support fish layers. It is commonly used in West African coastal towns but suffers from heat loss and low smoking capacity. Its major issues include inefficient fuel use, excessive fish handling, and difficulty controlling heat.

1.4 Rectangular/Square Metal Oven

This oven is constructed from 44-gallon oil drums joined into a rectangular or square shape. It uses iron rods and wire mesh to hold fish but radiates a lot of external heat. While durable, it shares the same disadvantages as mud ovens, such as heat loss and inefficient fuel use.

2. Modern Fish Smoking Technology

Modern fish smoking technologies use electricity, gas, or smart systems to improve efficiency and product quality. These methods reduce manual labor, allow better control of temperature and smoke, and produce more consistent results. By integrating features like automation, sensors, and mobile applications, modern systems ensure safer, faster, and higher quality smoked fish production.

2.1 Electric Fish Smoking Kiln

Electric smoking kilns use heating coils or electric burners to produce controlled heat and smoke. They provide consistent temperature, faster smoking, and a cleaner environment compared to traditional ovens. However, they require stable electricity and may be costly for small-scale users.

2.2 SMOFIM

IoT-based smoking systems use sensors to monitor temperature, humidity, and smoke levels in real-time. Data is transmitted to mobile devices, allowing operators to adjust settings remotely. This technology improves efficiency, ensures food safety, and maintains uniform product quality.

2.3 Microcontroller-Based Smoked Fish Machine

Researchers from Tarlac State University developed a microcontroller-based smoked fish machine that automates boiling, smoking, and drying. Using Arduino, sensors, and PID control, the system maintains precise temperatures for cooking and smoking to ensure consistent quality. Made with food-grade materials, it minimizes contamination, shortens processing time, and improves safety for operators.

The development of fish smoking technologies reflects a continuous effort to improve efficiency, safety, and product quality. Traditional methods, such as open-fire smoking, mud ovens, and metal drum ovens, have been effective for preservation but are often labor-intensive, fuel-inefficient, and inconsistent in output. These older systems also expose workers to health risks due to smoke and heat exposure, while producing unevenly smoked fish because of poor temperature control.

Improvements like the Chorkor oven, cylindrical metal oven, rectangular mud oven, and square metal oven introduced higher capacity and better fuel efficiency, marking a significant step toward modernization. However, even these semi-improved systems still rely heavily on manual operation and lack precise environmental control.

Modern technologies, such as electric fish smoking kilns, IoT-based systems (SMOFIM), and microcontroller-based machines, represent a leap forward by integrating automation, sensors, and electronic control systems. These advancements enable consistent temperature regulation, faster smoking times, and real-time monitoring through mobile devices—resulting in safer working conditions, reduced labor, and superior product quality.

3. Patents and Innovation

The development of smoking machines for fish, particularly Tinapa, represents a major step forward in food preservation and value-adding processes. Patents in this field have focused on improving efficiency, portability, and product quality while minimizing the drawbacks of traditional smoking methods, such as uneven heat distribution and undesirable burnt flavors. Innovations in this area include compact cabinet designs, controlled smoke generation, and eco-friendly heating mechanisms that improve the curing process and make it accessible to both commercial and small-scale producers.

3.1 Apparatus for Smoking Fish or Meat (Google Patent: US2789877A)

The patent entitled *"Apparatus for Smoking Fish or Meat"* by Fredrick H. Pfundt (1957) [27] introduces a portable and low-cost smoking cabinet designed for curing fish efficiently. The invention makes use of an enclosed metal chamber with screen trays where fish are placed, and an electric hot plate that heats a wood log to produce smoke through pyrolysis rather than direct combustion. This method provides cleaner smoke and imparts a more desirable flavor compared to traditional smokehouses that relied on burning wood. The cabinet also includes a chimney with a butterfly valve and draft regulator, allowing users to control airflow and smoke density during the smoking process. By combining portability, affordability, and improved smoking quality, this invention offered fishermen and small-scale producers a practical way to preserve fish such as Tinapa immediately after their catch.

3.2 Smoked Food (Google Patent: WO2015/007742A1)

This patented invention focuses on producing smoked food with reduced levels of harmful polycyclic aromatic hydrocarbons (PAHs) while keeping the smoky flavor intact.

The system works by using a selective filter, often made from zeolite minerals like clinoptilolite, to trap PAHs during or after the smoking process without removing the desirable taste compounds. Unlike traditional smoking methods, this approach prevents carcinogenic chemicals such as benzo[a]pyrene and chrysene from contaminating the food. The apparatus consists of a smoke generator, a chamber for smoking food, and a filter placed between them, making it adaptable to existing smoking setups. This technology not only makes smoked food safer for consumption but also supports the creation of liquid smoke flavorings and food additives with minimal toxic residues, aligning with international food safety standards.

3.3 Smokehouse for Fish (Google Patent: US2452016A)

The smokehouse for fish and other foodstuff, patented by Alfred L. Lyon in 1948, is an apparatus designed to produce a clean and sanitary smoked product by preventing ashes, embers, and cinders from contaminating the food. It features a specially constructed brick oven with a closed-in metal cylinder fireplace that allows only clean smoke to escape through vents near the top while trapping unwanted particles. Above the oven, there is a smoke room where fish, meat, or other food items are placed on shelves, wires, or racks to ensure even exposure to the smoke. Additional features include dampers, flue pipes, and fans that regulate airflow, smoke density, and heat distribution to achieve consistent smoking results. This innovation ensured higher quality smoked products, making them more appetizing, safe, and efficient to produce compared to traditional open-fire smoking methods.

The reviewed patents demonstrate a continuous advancement in fish smoking technology, emphasizing efficiency, safety, and product quality. Early innovations, such as Lyon's Smokehouse for Fish (US2452016A), focused on creating a cleaner and more controlled smoking environment by preventing contamination from ashes and embers while improving airflow and heat regulation. This marked a transition from open-fire methods to enclosed, sanitary systems.

Later developments, such as Pfundt's Apparatus for Smoking Fish or Meat (US2789877A), introduced portable and affordable designs that utilized electric heating and controlled smoke generation through pyrolysis. This not only produced cleaner smoke and better flavor but also made the technology accessible to small-scale fish processors.

More recent patents, like Smoked Food (WO2015/007742A1), highlight the integration of modern food safety and environmental considerations, introducing filters that reduce harmful compounds such as polycyclic aromatic hydrocarbons (PAHs) while preserving the desired smoky flavor.

Overall, these patents illustrate the progression from traditional, manual smoking systems to more advanced, hygienic, and controlled technologies. The innovations collectively aim to improve product quality, reduce health hazards, and promote sustainability. The SmokeFin Machine builds upon these principles by combining automation, real-time monitoring, and eco-friendly design—bringing traditional Tinapa production into efficient, and safe food processing.

Conceptual Literature

While traditional wood smoke imparts desirable sensory qualities to smoked fish, it also introduces harmful components, such as polycyclic aromatic hydrocarbons (PAHs). The growing preference for smoke preparations especially purified forms like liquid smoke reflects a shift toward safer, more controlled smoking methods that minimize health risks without compromising product quality. This presents an opportunity for innovation in smoke technology and regulatory development in smoked fish production [5]. The machine helps reduce the harmful effects of smoke by trapping it inside the chamber and releasing it safely through a chimney. Although some smoke still causes discomfort, it's a big improvement over traditional smoking method. The smoked fish comes out clean, hygienic, and with a pleasant smell [6]. The Smoked tool fish which effective, hygienic, and ecofriendly can increase the production of smoked fish and reduce environmental pollution from the smoke produced. This shows that smoked fish products using Smoked tool fish which effective, hygienic, and eco-friendly have water content according to the standards determined by SNI [7]. The improved fish smoking (saltation) equipment offers several benefits: it is ergonomically designed for user comfort and safety, provides efficient heating and smoke systems, and enables the production of durable, diverse tuna-based products [8].

The length of smoking depends on the desired flavor and moisture level of the smoked product, while the length of the cooking phase depends on the smoke temperature. In reference, the typical internal fish and oven temperatures during the smoking cycle can be used as a guideline. As the temperature within the smoking chamber increases slowly from the ambient temperature at the beginning of the smoking process to about 82.2 °C in 6 h, the internal temperature of fish also increases to about 60 °C. After about 6 to 8 h of smoking, the temperature within the smoking chamber must be increased to 85 °C or higher so that the internal temperature of fish reaches 68 to 72 °C. This peak cooking temperature should be held for at least 30 min to ensure that the fish is cooked thoroughly. Once the fish is cooked, the temperature can be decreased to about 40–50 °C to achieve the desired flavor and moisture level [9]. The heat and smoke source are one of the key factors that affect the hot fish smoking process. Coconut shell, one source that is available locally, is a biomass fuel that can be used because it has a high heating value and good smoke quality (lower ash content) [10].

A microcontroller-based smoked fish machine was developed to automate boiling, smoking, and drying using an Arduino system with PID temperature control. The machine maintains cooking temperatures of 75–90°C and smoking temperatures of 60–70°C. Built with food-grade materials, it ensures hygienic processing and completes the entire cycle in about 60 minutes [11]. Studies show that many producers still depend on family-taught skills and traditional methods. While these practices help preserve cultural identity, they often make it difficult to introduce standard procedures and modern innovations. Producers also struggle with issues such as weather disruptions, limited access to markets, and uneven product quality, especially since modern techniques and HACCP standards are not fully applied. To address these concerns, researchers suggest the need for consistent production processes, stronger quality control, and better food safety through proper training and improved facilities. They also recommend embracing modern technologies, exploring wider market opportunities, and adopting sustainable practices to help ensure the industry's long-term success [12].

Given the importance of the smoked fish industry in sustaining local economies, there is a growing need for modern and reliable smoking technologies. To address this, researchers developed an Internet of Things (IoT)-based monitoring and control system for fish smoking machines. The system was designed to manage temperature levels, regulate speed, and control machine components remotely. Findings revealed that IoT integration allows components such as blowers, servo motors, and lights to be operated remotely, while temperature sensors enable real-time monitoring of the smoking process through smartphones [13]. Using IoT in fish smoking machines makes the process more convenient and efficient, since operators can check and control the system in real time through their mobile phones. This not only saves time and effort by removing the need to be physically present, but also allows the data collected from the machine to be analyzed to improve productivity and ensure smoother production [14].

The SmokeFin machine is a modernized automatic smoking machine designed for Tinapa production. Unlike the traditional open-fire method, this system integrates Internet of Things (IoT) technology to enhance efficiency, safety, and product quality. By incorporating automation, SmokeFin reduces labor requirements, minimizes health risks from smoke exposure, and ensures more consistent outcomes in fish smoking. This innovation aims to revolutionize small-scale fish processing by providing a reliable, eco-friendly, and sustainable solution. The proposed system is composed of several key components that ensure a smooth and efficient smoking process. A heating element provides controlled temperature, while sensors monitor heat, humidity, and smoke levels in real time. A microcontroller coordinates the system, supported by wireless communication modules that allow remote monitoring and control through a smartphone. Actuators such as blowers and relays regulate airflow and smoking intensity, ensuring uniform product quality. Altogether, these elements form an integrated system that modernizes Tinapa production for both safety and scalability.

The development of SmokeFin has improved efficiency by automating critical tasks such as temperature regulation, airflow control, and smoking duration. The system reduces dependence on manual supervision, allowing fish processors to save time and energy while maintaining consistent product standards. Furthermore, the integration of IoT enables real-time monitoring, which empowers users to oversee operations remotely and make timely adjustments to ensure food safety and reliability. Beyond productivity, SmokeFin enhances product quality. The controlled smoking environment helps preserve the flavor and texture of Tinapa while meeting hygienic standards. With better control of moisture and smoke density, the system produces a cleaner, safer, and more appealing product. This contributes to greater consumer confidence and strengthens the competitiveness of local Tinapa in broader markets.

As technology advances, the SmokeFin machine has the potential to grow even smarter with new technologies. Possible improvements include using machine learning to automatically adjust smoking conditions, adding solar power to make the system more energy-efficient, and using cloud-based tools to track quality and ensure food safety. Still, there are challenges that may slow down its wider use, such as the high cost of starting up, the need to train users, and regular maintenance. Overcoming these hurdles will require teamwork among researchers, local communities, and policymakers to make the technology more practical, affordable, and sustainable for everyone.

Conceptual Framework

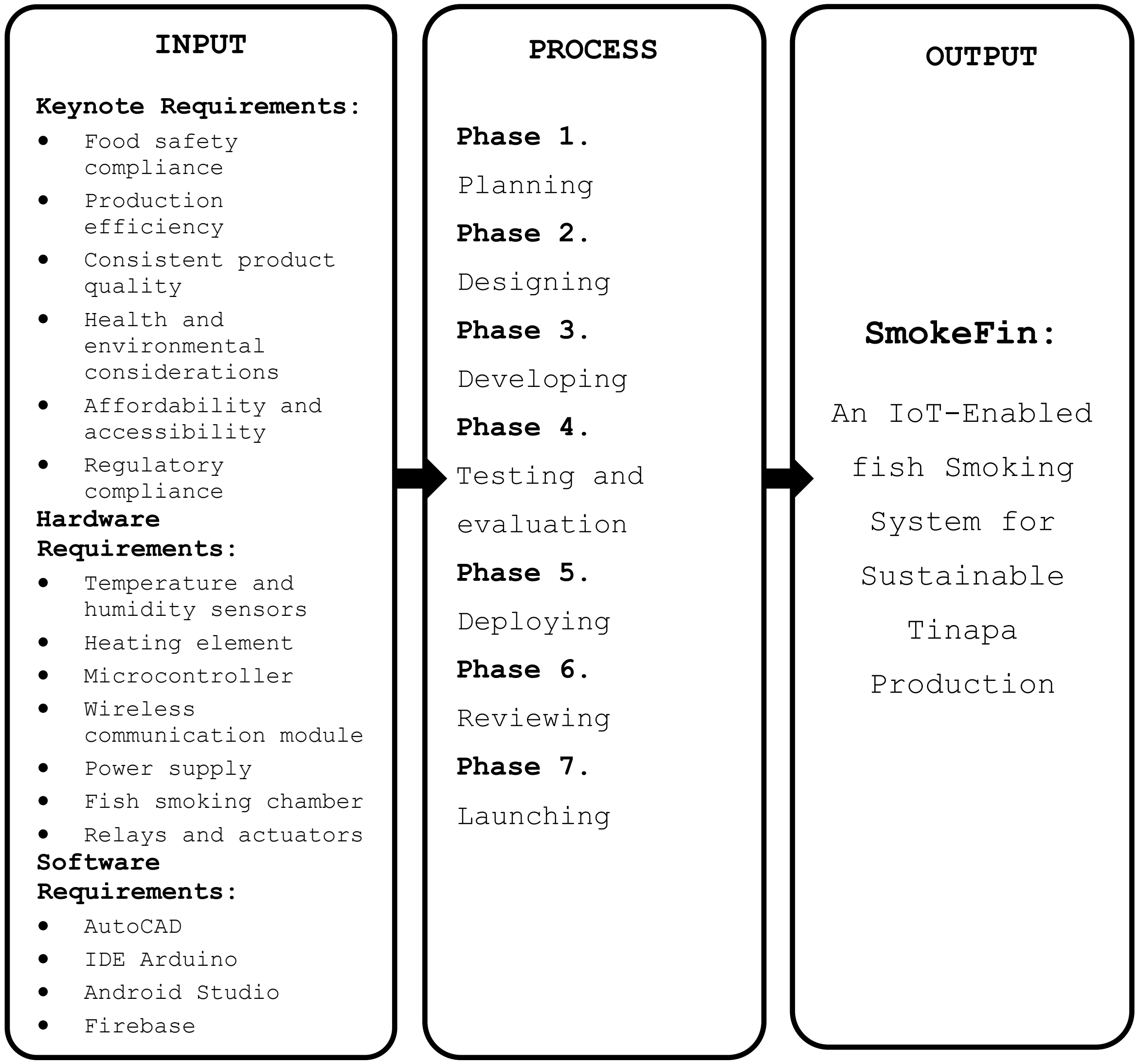


Figure 1. Conceptual Framework of the Study

Figure 1 presents the conceptual framework of the study, which is divided into three major parts: input, process, and output phases. The input phase provides the fundamental needs and resources required for the development of the project. It includes considerations such as food safety, production efficiency, consistent product quality, environmental and health factors, and regulatory compliance. Furthermore, it identifies the essential hardware components such as sensors, microcontrollers, heating elements, and wireless communication modules, as well as the software tools like Arduino IDE, cloud database, and mobile application needed to run the system. The process phase outlines the phases of system development. This involves planning, where the necessary materials are identified and prepared. The designing phase focuses on creating and refining the prototype of the IoT-based fish smoking system with input from experts and users. The developing phase where developers integrate hardware and software components into a working system, while the testing and evaluation phase ensures that the prototype is reliable, efficient, and capable of producing consistent results. In addition, the testing and evaluation stage ensures the system's reliability, efficiency, and consistency by conducting hardware and software tests, gathering user feedback on usability and satisfaction, and identifying issues for improvement in future iterations. The output phase represents the final product of the study: SmokeFin, an IoT-enabled fish smoking system for sustainable Tinapa production. This system will improve the smoking process, offers a modern, and user-friendly solution to traditional fish smoking practices, supporting small-scale fish vendors and local communities.

Multiple Design Constraints and Tradeoff

In the development of the SmokeFin system, several constraints emerged that affected the design process and implementation. These constraints needed careful consideration and balancing to ensure that the system remained functional, safe, and practical for both small-scale processors and consumers.

Cost Constraints

Cost was one of the primary challenges encountered in the development of the SmokeFin system. The integration of sensors, microcontrollers, wireless modules, and actuators required considerable financial investment. Some of these components were not available locally and had to be sourced externally, which increased procurement and shipping expenses. Since the project aimed to create an affordable and accessible solution for small-scale fish vendors, the budget had to be carefully managed. Tradeoff was necessary between selecting high-performance parts and maintaining affordability to keep the technology practical for end users.

Time Constraints

Time also is one of significant limitations during the project’s development. Building an IoT-based system required multiple phases—design, hardware assembly, software integration, testing, and refinement—all of which demanded careful coordination. Delays in the availability of certain electronic components, as well as extended testing for temperature and humidity accuracy, added to the overall project timeline.

Health and Safety Constraints

Because the project involved food production, strict attention was given to health and safety. The smoke box and components needed to be food-grade and safe to prevent contamination of the Tinapa. Moreover, since traditional fish smoking exposes processors to harmful smoke, the design had to ensure safe ventilation and minimal user exposure to hazards. The system was tested to confirm compliance with basic food safety standards, with trade-offs made to balance performance, durability, and hygiene.

Environmental and Sustainability Constraints

Another key consideration was the environmental impact of the smoking process. While the system minimized direct smoke exposure, it still required fuel or energy sources to operate. To make the system more sustainable, the group planned to use coconut by-products or residues as the main fuel source. These materials are abundant, low-cost, and renewable, making them an eco-friendly alternative to traditional fuel. Using coconut shells and husks also helps reduce agricultural waste, turning it into a useful resource for fish smoking. This approach supports both environmental protection and cost efficiency, while still providing the necessary heat and smoke quality for the process.

Technical Constraints

The SmokeFin system relied heavily on IoT technologies, which required stable connectivity and precise calibration of sensors. Limitations in local internet availability in some fishing communities presented a challenge for real-time monitoring. Additionally, maintaining sensor accuracy in a high-heat, high-humidity environment was complex.

Chapter III

PROJECT DESIGN AND METHODS

Project Analysis

Fish smoking has long been an important preservation method in the Philippines, particularly in coastal and riverside communities where fish serves as both a primary food source and a livelihood. However, traditional fish smoking methods are labor-intensive, health-risking, and often yield inconsistent product quality due to uneven temperature and smoke control. As demand for safer and more efficient food production continues to grow, there is a pressing need to modernize Tinapa (smoked fish) processing with innovative technologies.

The SmokeFin project was chosen to address the longstanding challenges of traditional fish smoking methods, which are labor-intensive, inconsistent, and hazardous to the health of small-scale processors. The primary goal of this project is to design and implement an IoT-enabled system that automates and monitors the Tinapa smoking process, ensuring improved efficiency, safety, and product quality. By integrating sensors, a microcontroller, and wireless communication, the system provides real-time data on temperature, humidity, and duration, allowing users to remotely monitor the smoking process. This methodology is significant because it combines food safety compliance, technological innovation, and sustainability to meet the pressing needs of fish-based communities. The chapter outlines the project objectives, the research foundation that guided the design, the conceptualization of the solution, and the architecture of the SmokeFin system.

The problem definition highlights the need to modernize fish smoking practices, particularly in communities where Tinapa is a major livelihood. Traditional methods expose processors to harmful smoke and produce inconsistent results, leading to inefficiency and health risks. The SmokeFin project seeks to overcome these problems by creating an automated system that is affordable, user-friendly, and suitable for small-scale production. The design must meet key requirements such as accurate temperature and humidity monitoring, safe handling of smoke, food safety compliance, and energy efficiency. Constraints include cost limitations, environmental considerations, and the need for reliable connectivity, which were carefully reviewed and adjusted during the course of the project to refine the final set of system requirements.

A comprehensive background study was conducted to examine existing technologies and innovations in fish smoking. Traditional ovens such as the Chorkor and cylindrical metal ovens were analyzed alongside modern systems like electric kilns and IoT-enabled prototypes. Literature reviews also revealed gaps in consistency, sustainability, and scalability, which the SmokeFin project aims to address. Insights from existing research—such as the benefits of microcontroller-based automation and the potential of IoT for real-time monitoring—played a crucial role in shaping the system design. This grounding ensured that SmokeFin not only improves on traditional methods but also aligns with international food safety and environmental standards.

The design conceptualization process involved brainstorming and evaluating different approaches to integrating sensors, actuators, and microcontrollers. The final design was chosen for its practicality, innovation, and close alignment with project goals. By automating heat control, smoke circulation, and monitoring, while also enabling user-friendly remote access, the selected concept balanced technical sophistication with affordability and ease of use for small-scale vendors.

The system architecture integrates several key components into a cohesive framework. Sensors such as K-Type Thermocouple sensor to collect environmental and process data from the smoking chamber. These inputs are processed by the

ESP32 microcontroller, which controls actuators like the heating element, fan, and servo damper to regulate conditions inside the chamber. Safety mechanisms, including overtemperature cutoff and fire detection, add reliability to the process. Wireless connectivity ensures that data is transmitted and accessed through mobile applications, giving users full visibility and control. The system is powered by AC mains with battery backup and energy management components, ensuring stable and continuous operation. Altogether, the architecture demonstrates how IoT technology can transform traditional Tinapa production into a safer, and more sustainable process.

Block Diagram

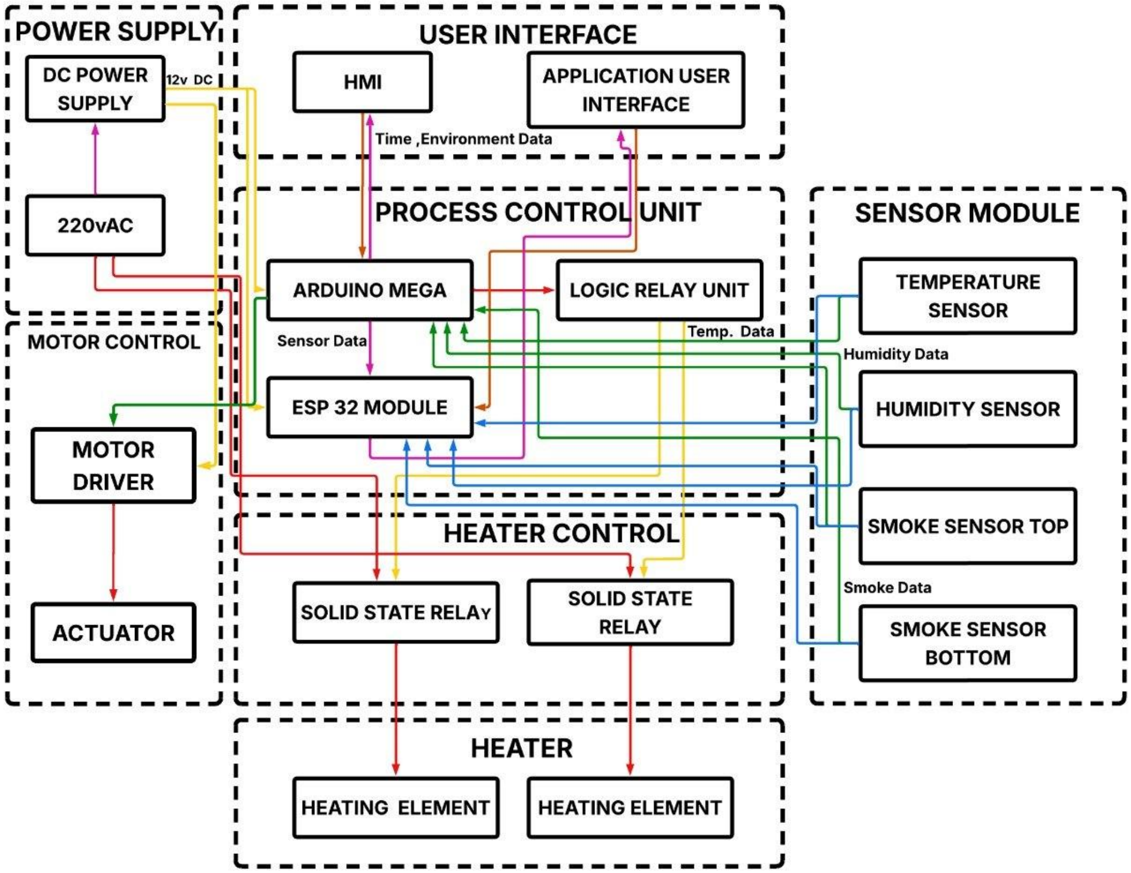


Figure 2. Block Diagram of SmokeFin

Figure 2 shows the block diagram of the SmokeFin System. The system is composed of several major components including the power supply unit, user interface, process control unit, sensor module, motor control, and heater control. These components work together to automate and monitor the fish smoking process efficiently. The operation of the system begins at the power supply section, where a 220VAC source is converted into 12V DC power through the DC power supply. This provides the necessary electrical power for the Arduino Mega, ESP32 module, sensors, relays, and other electronic components of the system. The process control unit serves as the central controller of the system. The Arduino Mega receives input data from the different sensors such as the temperature sensor, humidity sensor, smoke sensor top, and smoke sensor bottom. These sensors continuously monitor the environmental conditions inside the smoking chamber and send real-time data to the controller for processing and analysis.

The ESP32 module is integrated into the system to provide wireless communication and data transmission. It communicates with the Arduino Mega and transfers sensor readings and system status to the application user interface. This allows the user to monitor the operation of the SmokeFin System remotely through a mobile or web-based application. The user interface section consists of the Human Machine Interface (HMI) and the application user interface. These interfaces display important information such as temperature, humidity, smoke level, and operating time. Through these interfaces, the user can monitor the status of the system and interact with the controls when necessary.

For the heater control section, the Arduino Mega sends control signals to the logic relay unit and solid-state relays. The solid-state relays regulate the operation of the heating elements, which are responsible for maintaining the required temperature inside the smoking chamber. This ensures stable and controlled heating during the smoking process. The motor control section includes the motor driver and actuator. The motor driver receives commands from the process control unit and controls the actuator's movement or operation. This mechanism supports the automated functions of the SmokeFin System. Overall, the block diagram illustrates how the SmokeFin System integrates sensing, monitoring, wireless communication, heating control, and automation technologies to achieve an efficient and controlled fish smoking process.

Flowchart

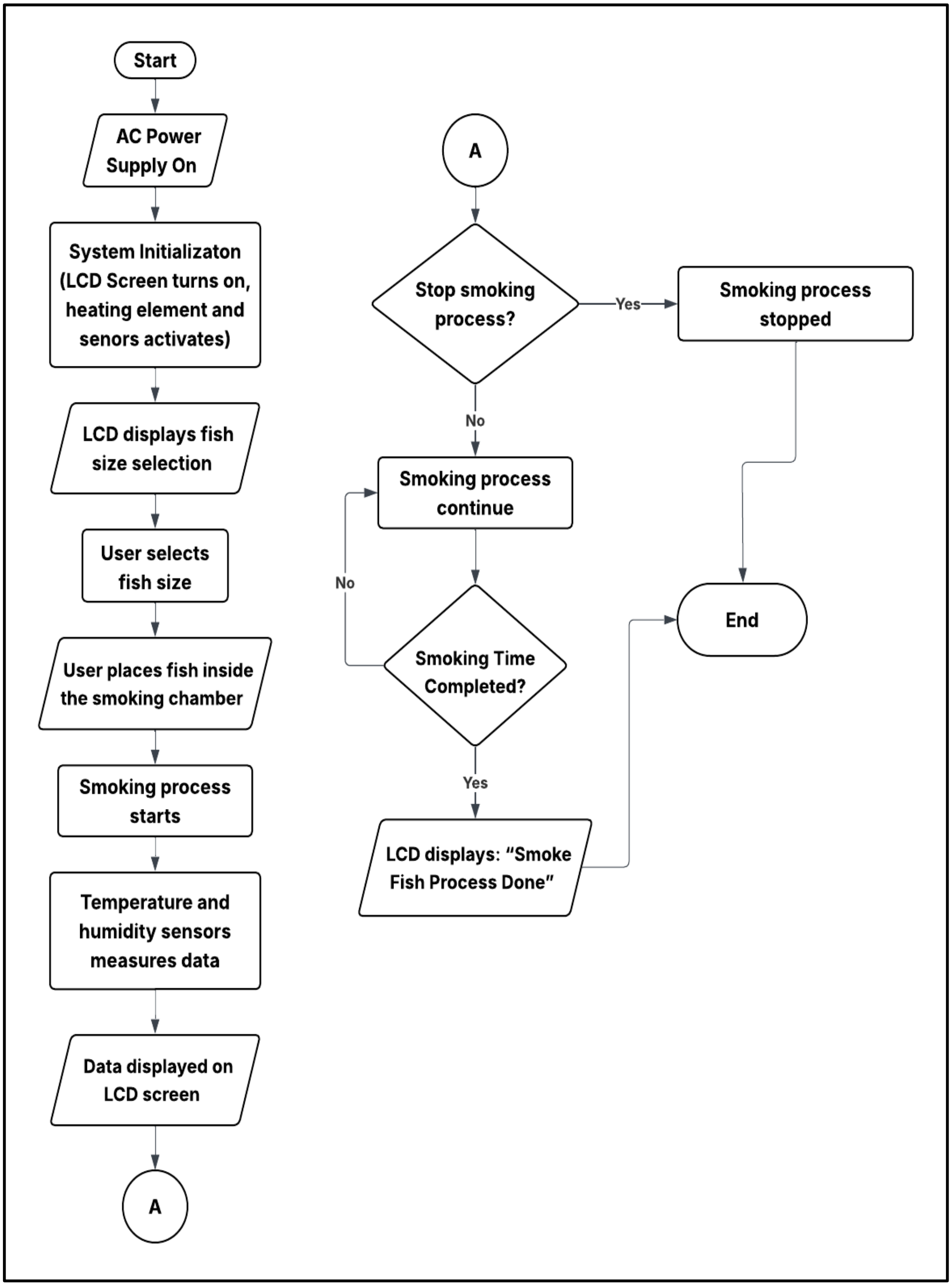


Figure 3. Flowchart of SmokeFin System

Figure 3 shows the flowchart of SmokeFin System. The process begins when the system is powered on using an AC power supply, which initiates system initialization. During initialization, the LCD screen turns on, and the heating element along with the temperature and humidity sensors are activated.

Once initialized, the LCD displays fish size options for the user. After the user selects the desired fish size, the fish is placed inside the smoking chamber, and the smoking process begins automatically. As the process runs, the temperature and humidity sensors continuously measure environmental data inside the chamber, which is then displayed on the LCD screen for real-time monitoring.

In addition, the user has a choice when they want to stop the smoking process even if the time is not over. This allows the user to stop the smoking process depends on their prefer doneness of the smoke tinapa. Once the smoking time is finished, the LCD displays the message "Smoke Fish Process Done", indicating successful completion of the operation, and the system then proceeds to the end of the process

.

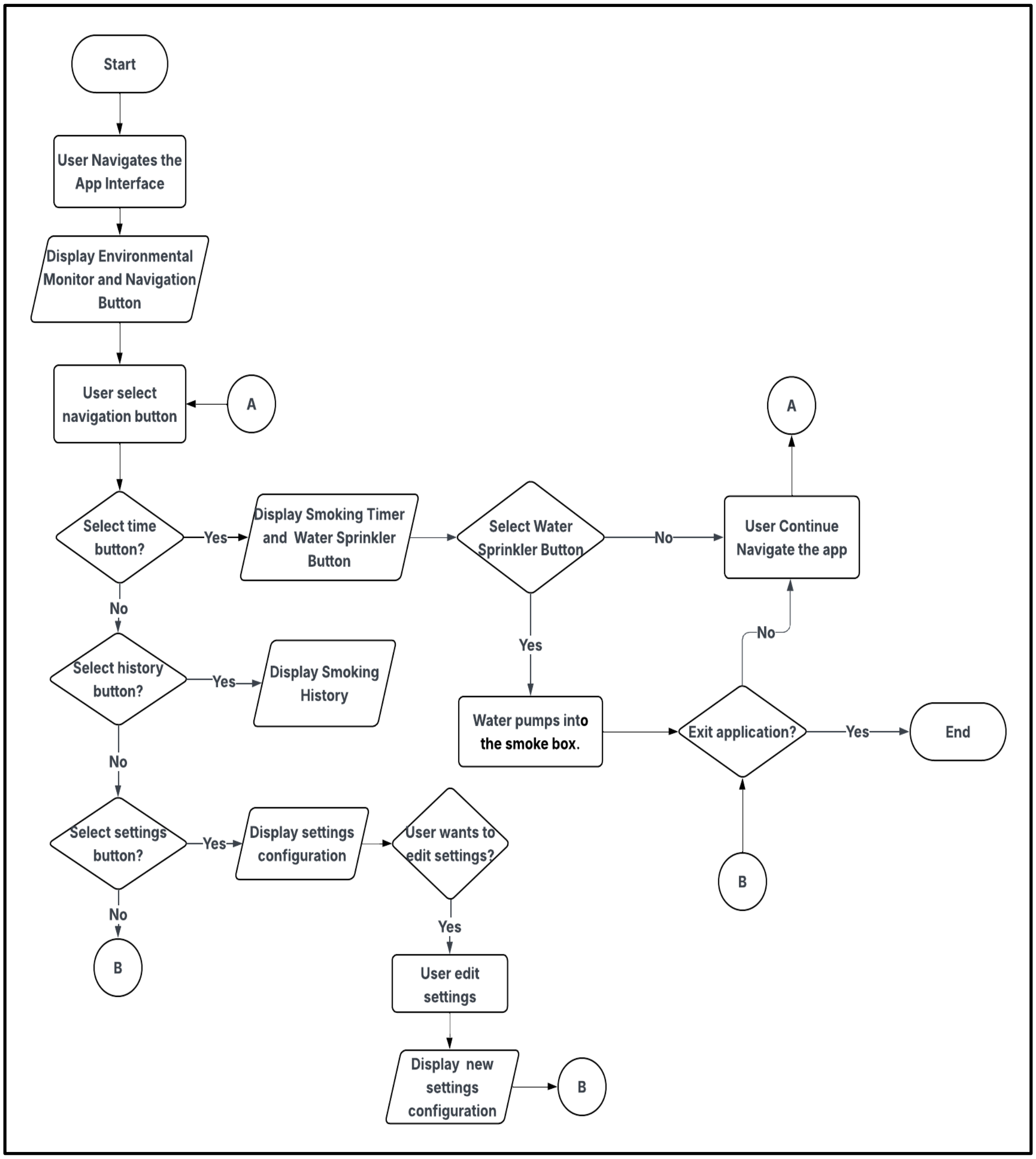


Figure 4. Flowchart of the Application of SmokeFin

Figure 4 shows the flowchart of the mobile application of the SmokeFin. The process starts with the user navigating the app interface. The application displays an environmental monitor and navigation button. The application presents options to select a time button, a history button, or a settings button. If the user selects the time button, the application displays a smoking timer and an emergency button. Additionally, if the user selects the water sprinkler button, the water sprinkler will wet the smoke dust in smokebox and the smoking process will stop.

Moreover, if the user selects the history button, the application displays the smoking history. If the user selects the settings button, the application displays the settings configuration. The user can then choose to edit the settings. If the user edits the settings, the application displays the new settings configuration. Finally, the user can choose to exit the application. If the user exits the application, the process ends. Otherwise, the user continues navigating the app.

Project Management

The development of the SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production required careful planning, coordination, and resource management to ensure its successful completion. Since the project involved the integration of mechanical components, electronic modules, and IoT technology, the proponents faced the challenge of balancing technical complexities with practical usability. Effective communication and collaboration among team members were essential in addressing design issues, coordinating tasks, and making timely decisions throughout the development process. Limited availability of electronic components and food-grade materials also demanded early planning, proper budgeting, and sourcing strategies to prevent delays.

Furthermore, the rising costs of equipment and transportation added financial pressure, requiring the team to manage resources wisely. Despite these challenges, the proponents' cooperation, adaptability, and commitment enabled the project to move forward, ensuring that each stage—from design to testing—was systematically executed to achieve the intended outcomes.

Methodology of the Project

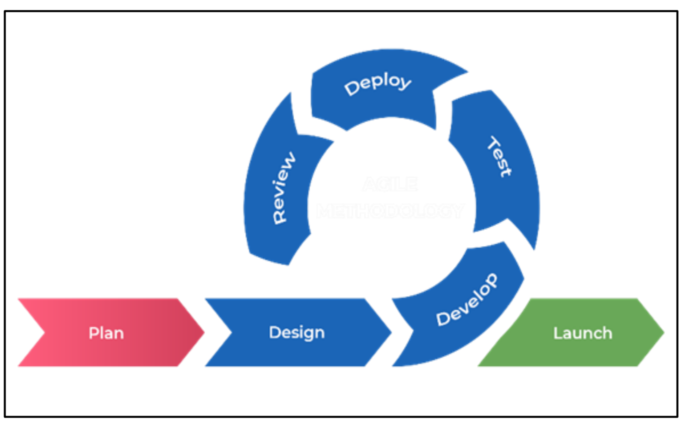


Figure 5. Agile Methodology

For the development of the hardware of the SmokeFin system, the developers decided to use Agile Methodology, which provided an iterative and flexible framework suited for integrating mechanical, electrical, and IoT components. Through multiple development sprints, the team systematically worked on hardware assembly, sensor calibration, mobile application integration, and heating and ventilation control. Each sprint included testing and validation, followed by refinements based on feedback from fish processors, technical experts, and advisers to ensure that the system addressed both technical objectives and user needs. This iterative process allowed the researchers to quickly respond to challenges such as connectivity limitations, sensor accuracy, and food safety considerations, ensuring continuous improvement of the prototype. By applying Agile, the project achieved a reliable, user-friendly, and sustainable system that modernizes traditional Tinapa production through automation, real-time monitoring, and improved food quality.

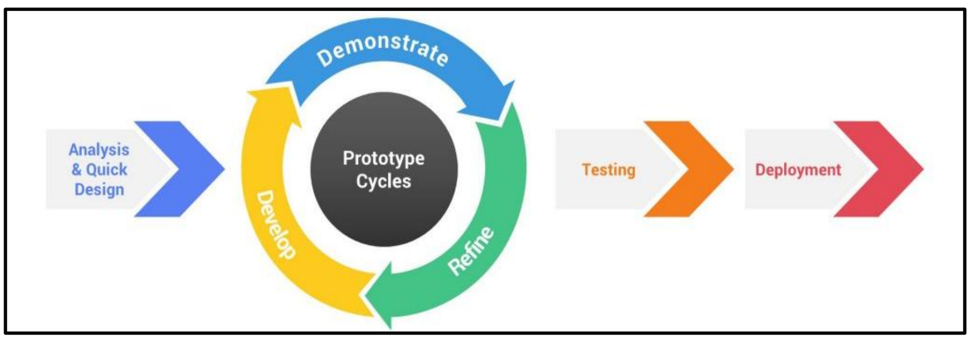


Figure 6. Rapid Application Development

For the development of the SmokeFin mobile application, the developers decided to use Rapid Application Development Methodology. The process starts with Analysis and Quick

Design, where the developers identified the core requirements for real-time monitoring of temperature, humidity, and smoke levels. During this phase, flowcharts and conceptual blueprints were rapidly created to map out how the app would interface with the ESP32 microcontroller and the Firebase database. This led directly into the Develop stage, or construction, where the actual coding in Android Studio and the wiring of hardware components took place. The developers focused on creating a working version of the app as quickly as possible to ensure that the data transmission between the smoke machine and the user's mobile device was functional. Once a prototype was ready, the project moved into the Demonstrate and Refine phase. The system was shown and tested in simulated environments to see if the interface was intuitive and if the remote-control features worked as intended. Feedback gathered during these demonstrations allowed the team to immediately refine the software, fixing UI bugs or adjusting sensor calibration logic. This was followed by testing phase to ensure compliance with quality standards like ISO 25010:2023, verifying that the app was reliable and secure. Finally, the deployment phase saw the transition of the SmokeFin system from a lab prototype to a practical tool for the beneficiaries, providing a modernized and safer fish-smoking process through the finalized E-Tinapa application.

Requirement Specifications

The requirement specification for SmokeFin focuses on developing a safe, efficient, and sustainable IoT-enabled fish smoking system that modernizes the traditional Tinapa-making process. The system automates critical tasks such as heat regulation, smoke circulation, and humidity monitoring through a microcontroller integrated with sensors, actuators, and wireless communication modules. Its purpose is to ensure consistent product quality, improve production efficiency, and reduce health risks associated with manual smoking. The prototype must be cost-effective, user-friendly, and designed for small-scale fish vendors, while being scalable for future enhancements such as solar integration or machine learning-based optimization. Furthermore, the system must comply with food safety and environmental standards, emphasizing sustainability, affordability, and safe consumer products.

Knowledge Requirements

The knowledge requirements for this project include a solid foundation in embedded systems, particularly in microcontroller programming (ESP32/Arduino) to integrate sensors, actuators, and wireless modules for automation and monitoring. The proponents must understand mechanical design principles for constructing a safe, food-grade smoking chamber with proper ventilation and insulation. Electrical system knowledge is also essential for safely connecting heating elements, blowers, relays, and protection devices such as MCBs and RCBOs. In addition, expertise in software development is required to design a mobile application using Android Studio for real-time monitoring and remote control of the system. Knowledge of AutoCAD is needed for technical design and layout visualization, while familiarity with IoT protocols ensures secure and reliable data transmission. Finally, awareness of project management, food safety standards (ISO 22000), and sustainability practices is necessary to deliver an affordable, ethical, and environmentally friendly solution.

Software Requirements

Other crucial tools supporting the technical control and design visualization of the system of SmokeFin are its software requirements. These software applications are essential in creating the system's design, developing the mobile application for monitoring, and LCD interface to display real-time data and system status for the users.

Figure 7. Arduino IDE

Arduino IDE serves as the primary platform for coding and uploading the control logic to the ESP32 microcontroller, which acts as the brain of the SmokeFin system. Through this software, the program that handles sensor inputs such as temperature, humidity, and smoke levels and actuator outputs such as heating element, and fan is written, compiled, and tested. The IDE also provides libraries that simplify the integration of multiple sensors and wireless communication modules, ensuring accurate real-time monitoring and automation.

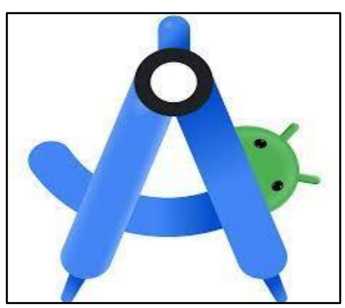


Figure 8. Android Studio

Android Studio is used to develop the mobile application that serves as the user interface for monitoring and controlling the SmokeFin system. Through this platform, the app is programmed to receive real-time data from the ESP32 via Wi-Fi, displaying parameters such as temperature, humidity, and smoking duration. It also allows users to input desired settings, adjust conditions, and turn the system on or off remotely. Android Studio ensures that the app is user-friendly, reliable, and fully compatible with smartphones, providing convenience and accessibility for fish processors.

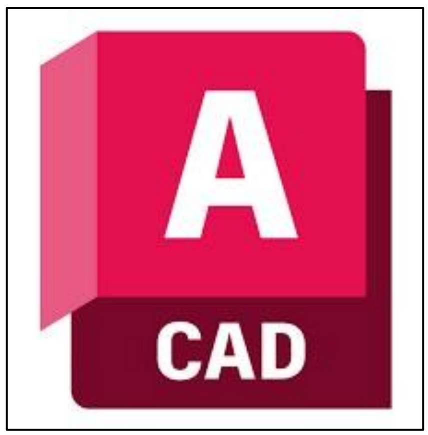


Figure 9. AutoCAD

AutoCAD is used to design and visualize the physical structure of the SmokeFin system, particularly the smoking chamber and placement of its components. This software allows the proponents to create precise technical drawings and layouts for the arrangement of sensors, heating elements, and housing.

Figure 10. Firebase

Firebase plays an important role in building the SmokeFin system by serving as the backend platform that enables real-time communication between the machine and the mobile application. Through its real-time database, it can store and update such as temperature, humidity, and cooking time, ensuring that users receive live updates on their devices. Firebase also provides. Firebase makes the SmokeFin system more reliable, efficient, and user-friendly.

Integration and Interoperability

The initiation of the SmokeFin system's development involves establishing clear objectives, scope, and requirements to modernize traditional Tinapa production through IoT technology. This requires consultations with small-scale fish processors and community stakeholders to ensure the system aligns with their needs in terms of safety, efficiency, and affordability. A feasibility study is carried out to confirm the integration of components such as sensors, microcontrollers, heating elements, and blowers, ensuring that each hardware element functions seamlessly within the IoT-enabled platform.

Following the design stage, complete system layouts and schematics are prepared to guarantee compatibility between hardware and software. AutoCAD is used to design the smoking chamber and component placement, while Arduino IDE and Android Studio support software development and simulations. This iterative design process incorporates expert and user feedback to refine both the hardware and application for optimal performance.

During the construction phase, the components—such as the ESP32 microcontroller, relays, thermocouples, heating system, and airflow mechanisms—are assembled into a working prototype. Simultaneously, the software is programmed to handle automation, monitoring, and wireless communication, ensuring smooth interaction between the mechanical functions and digital controls. Integration testing validates that the sensors, actuators, and mobile application operate reliably under real conditions.

In the implementation stage, the system is deployed for field testing, where smoked fish is produced under monitored conditions. Training sessions and user manuals are provided to assist operators in managing the automated process, while documentation ensures safe use and proper maintenance. Finally, the development team continues to evaluate performance and apply updates, using user feedback to enhance system reliability, efficiency, and interoperability with future technological upgrades.

Project Design and Description

This section discusses and presents the references and bases for the design of the SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production.

Design Details and Implementation

The SmokeFin will be implemented as an IoT-enabled fish smoking system that improves the tinapa smoking process while monitoring temperature, humidity, and smoke levels inside the chamber. These environmental conditions will be detected by sensors and regulated by a microcontroller unit, which controls the heating element, and ventilation system to maintain consistent smoking conditions. The system will also include a wireless communication module that sends real-time data to a mobile application for remote monitoring and control. Through the application, users can track the smoking process, receive notifications, and adjust settings as needed. This design improves product consistency, reduces manual labor, and promotes a more efficient and sustainable method of producing tinapa.

Wiring Diagram

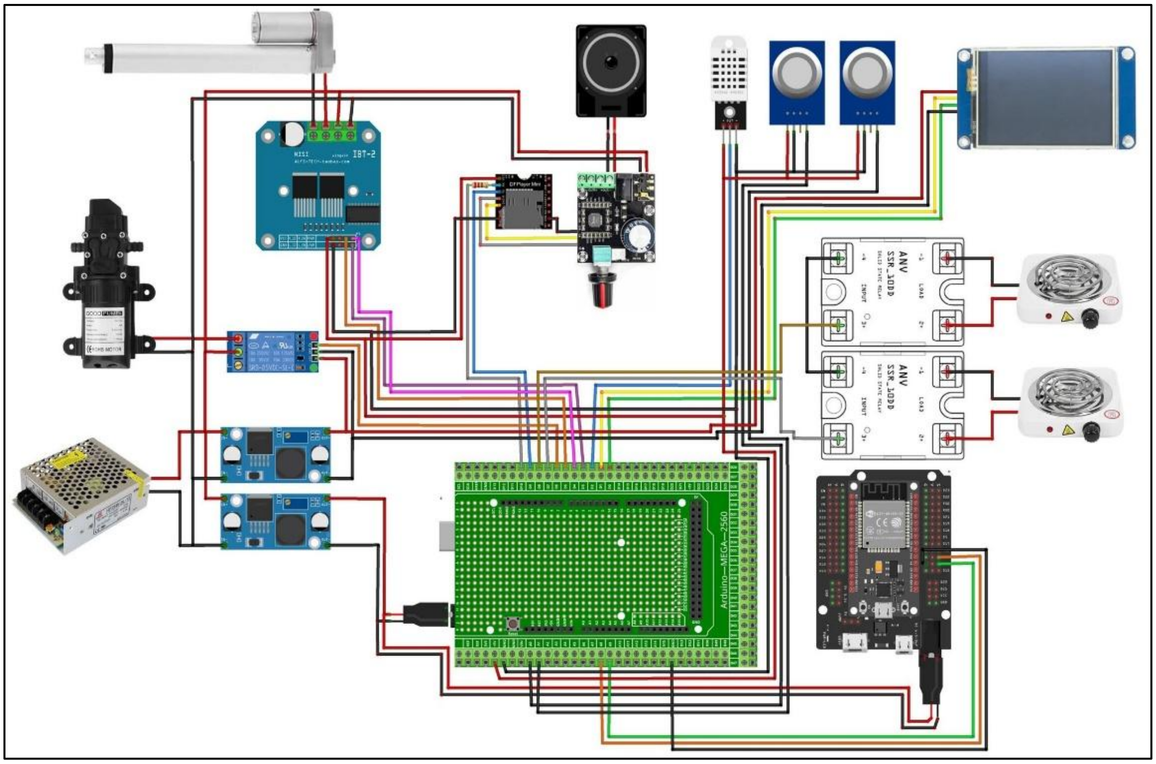


Figure 11. Wiring Diagram of SmokeFin

Figure 11 shows the wiring diagram that illustrates the complete hardware integration of the SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production. At the center of the system is the Arduino Mega 2560, which serves as the main controller responsible for processing sensor data and controlling all connected actuators. Environmental monitoring is achieved using a temperature and humidity sensor, which continuously measures the internal conditions of the smoking chamber. These readings are sent to the Arduino

Mega and are used to regulate the heating process to ensure consistent and safe smoking temperatures. Two gas or smoke sensors are also connected to detect smoke concentration, enhancing process control and safety.

In addition, heating control is managed through Solid State Relays (SSR) that interface between the Arduino and the electric heating coils. The SSRs allow the low-voltage control signals from the microcontroller to safely switch high-power heating elements on and off. This setup ensures electrical isolation and stable operation during prolonged smoking sessions.

Moreover, a DC motor driver module (IBT-2) is connected to a motor mechanism responsible for mechanical operations such as airflow control or chamber movement. The motor driver enables bidirectional control and sufficient current handling for heavy loads. A relay module is also used to switch auxiliary components such as the water pump, which helps regulate humidity or extinguish embers when necessary.

Furthermore, Power management is handled by a switching power supply, which converts AC mains voltage into DC voltage suitable for the system. Buck converter modules are used to step down voltage levels to meet the requirements of different components, ensuring stable and efficient power distribution throughout the system. User interaction and feedback are provided through an OLED display, which shows real-time system information such as temperature, humidity, and system status. An audio module with a speaker is integrated to deliver audible alerts or notifications during operation, such as when optimal smoking conditions are reached or when faults are detected.

Lastly, for connectivity, an ESP32 module is included to enable wireless communication. This allows the SmokeFin system to transmit data to a mobile application or cloud platform for remote monitoring and control, supporting its IoT functionality.

Perspective Design

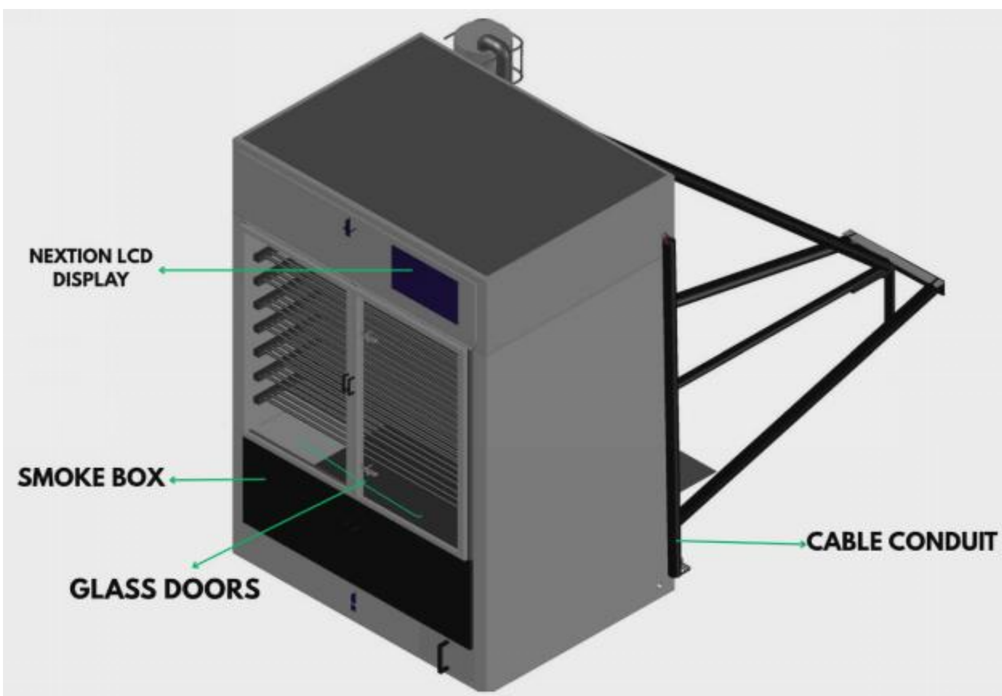


Figure 12. Overview of the SmokeFin Machine

Figure 12 illustrates the Overview of the SmokeFin Machine, showing its overall structural design and key external components. The system consists of a rectangular enclosed smoking chamber designed to provide controlled and uniform heat distribution during operation. At the front panel, a Nextion LCD display serves as the primary user interface, allowing real-time monitoring of parameters such as temperature, humidity, and system status. The chamber is fitted with glass doors that enable visual inspection of the fish without opening the unit, helping maintain stable internal conditions and reduce heat loss. Inside, multiple racks are arranged to accommodate batch processing of fish.

A smoke box is positioned at the lower section of the machine, functioning as the source of heat and smoke required for the smoking process. Additionally, a cable conduit is installed along the side to organize and protect electrical wiring, ensuring safe and reliable operation. Overall, the design emphasizes efficiency, safety, and ease of use for small-scale Tinapa production.

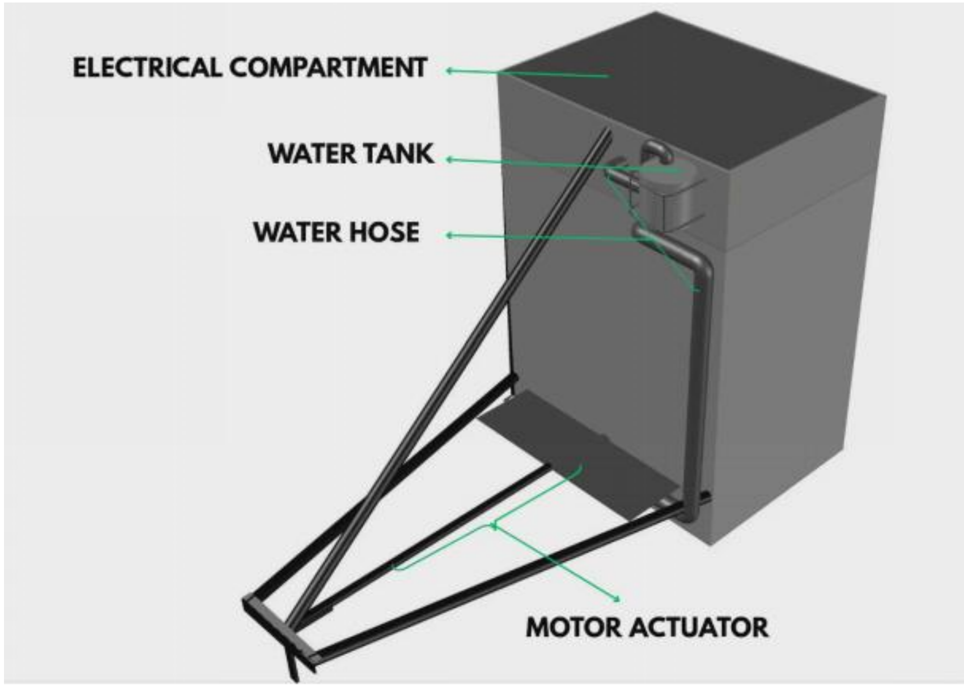


Figure 13. Left-Rear View of the SmokeFin Machine

Figure 13 presents the left-rear view of the SmokeFin Machine, highlighting its auxiliary components and support systems. At the upper section, the electrical compartment houses the machine's control circuitry and power distribution components, ensuring safe and organized electrical operation. Adjacent to it is the water tank, which serves as the source of water used during operation. A water hose is connected to the system and is utilized to extinguish the wood dust inside the smoke box when needed, providing an added safety mechanism to control or stop the smoking process. At the rear base, a motor actuator is integrated into the support frame, which functions to open or close the smoke box passage, allowing regulation or blockage of smoke entering the fish racks. The structural frame supports the entire assembly, enhancing stability and durability during operation. Overall, this view emphasizes the integration of electrical, safety, and mechanical control components that contribute to the efficient and controlled performance of the SmokeFin Machine.

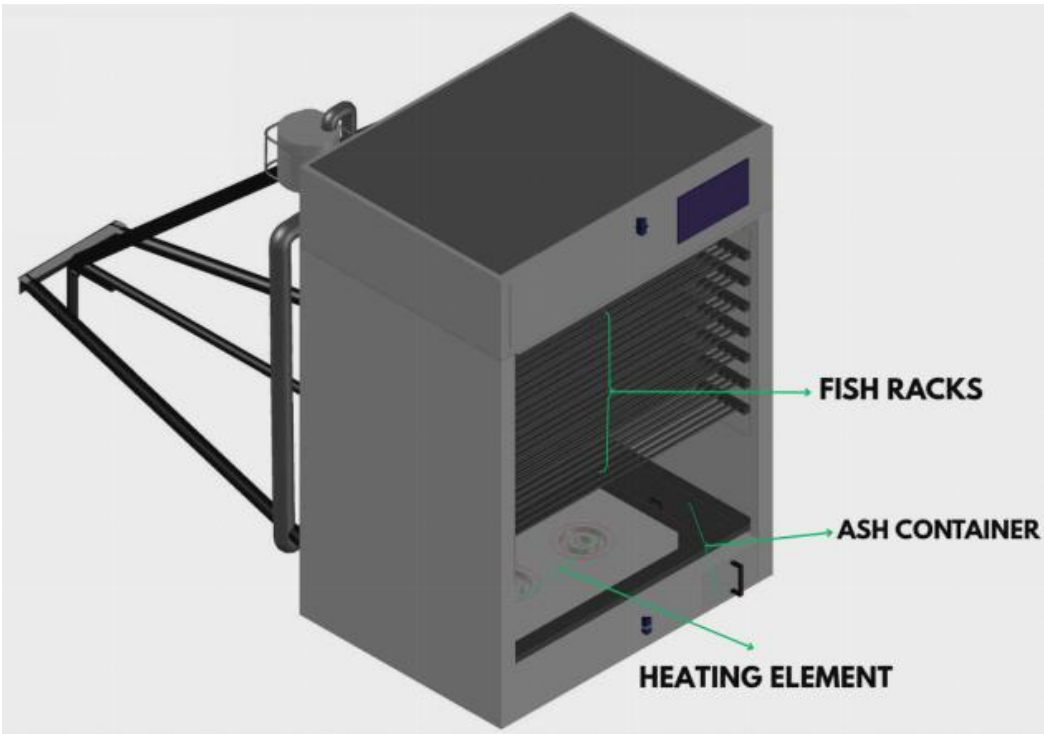


Figure 14. Open View of the SmokeFin Machine

Figure 14 shows the Open View of the SmokeFin System. It shows the interior features such as several layers of fish racks to maximize the capacity of each smoking session, and an integrated ash container sits at the base to manage byproduct waste. A heating element is positioned at the bottom of the unit to fuel the wood dust.

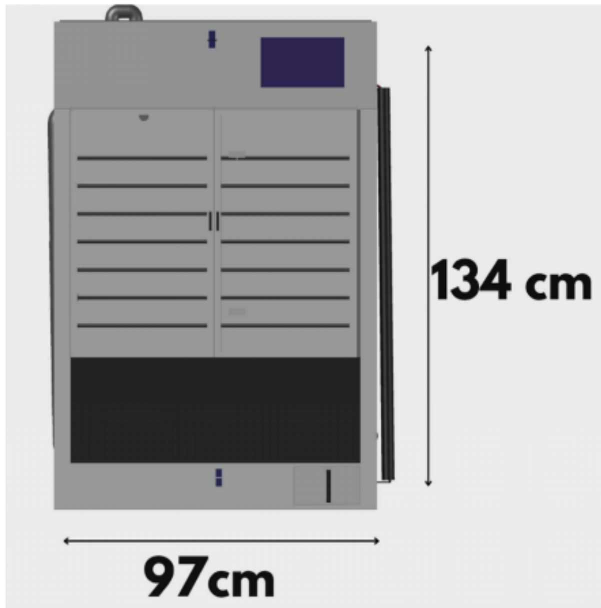


Figure 15. Front View of the SmokeFin Machine

Figure 15 presents the front view of the developed machine, highlighting its overall dimensions and external layout. The unit measures 134 cm in height and 97 cm in width, providing a compact yet functional structure suitable for operational use.

Graphical User Interface of the Mobile Application

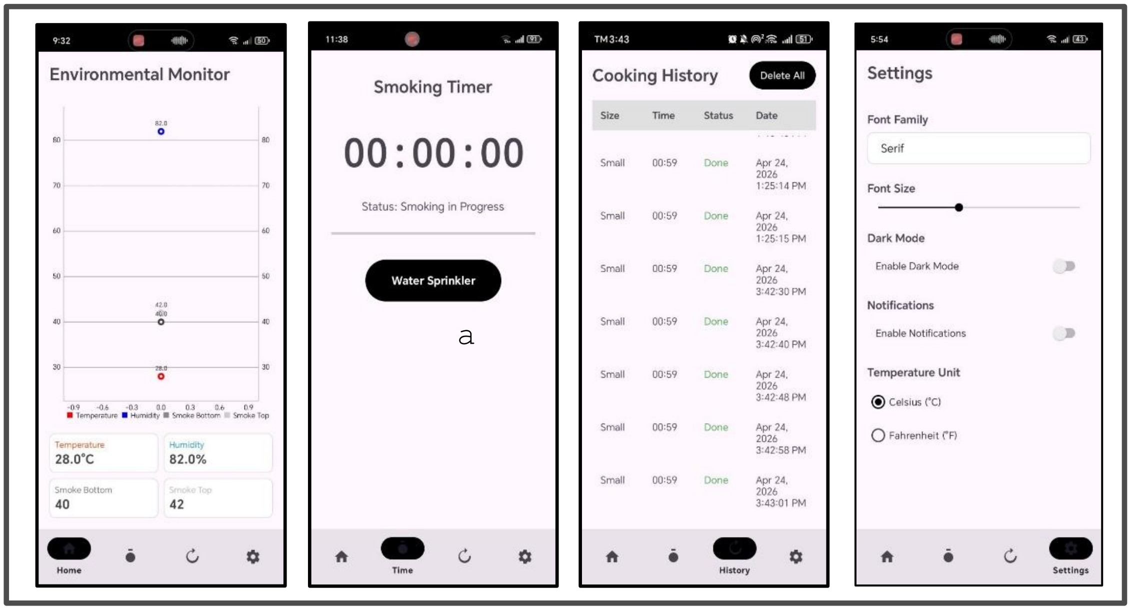


Figure 16. GUI of the Application of SmokeFin

Figure 16 shows the graphical user interface of the SmokeFin Mobile Application. It is a mobile-based interface designed to monitor, control, and ensure the safe operation of the fish smoking system. The application provides real-time cooking status, timer monitoring, and historical records of smoking sessions, allowing users to manage the tinapa production process efficiently and conveniently.

The Environmental Monitor screen displays real-time readings of temperature, humidity, and smoke levels inside the SmokeFin Machine during operation. Temperature and humidity values help ensure proper smoking conditions, while smoke sensors at the top and bottom monitor even smoke distribution within the chamber. A visual graph provides quick assessment of each parameter's status, supported by numerical data cards for accuracy. This feature ensures consistent tinapa quality, efficient operation, and safe smoking conditions.

Additionally, Cooking Timer screen displays the active smoking operation. It shows the current cooking time in minutes and seconds, along with a status indicator labeled *"Smoking in Progress"* to inform the user that the machine is currently operating. A Water Sprinkler button is included to allow user to wet the smoke dust in smokebox when the smoking process is already done in that day, so that the wood dust in the smoke box will not continue to rot, that it can be dried and use again next time.

Moreover, Cooking History screen records all completed and ongoing smoking sessions. Each entry includes the selected fish size, elapsed cooking time, current or final status, and the exact date and time the process started. This feature enables users to track previous operations, evaluate consistency in cooking results, and maintain proper documentation for quality control.

Lastly, settings screen allows users to customize the application's appearance and behavior. Users can select the font family, adjust the font size, and enable or disable dark mode. Notification settings can be toggled on or off, and the preferred temperature unit can be set to Celsius or Fahrenheit. These options provide a personalized experience for interacting with the SmokeFin machine.

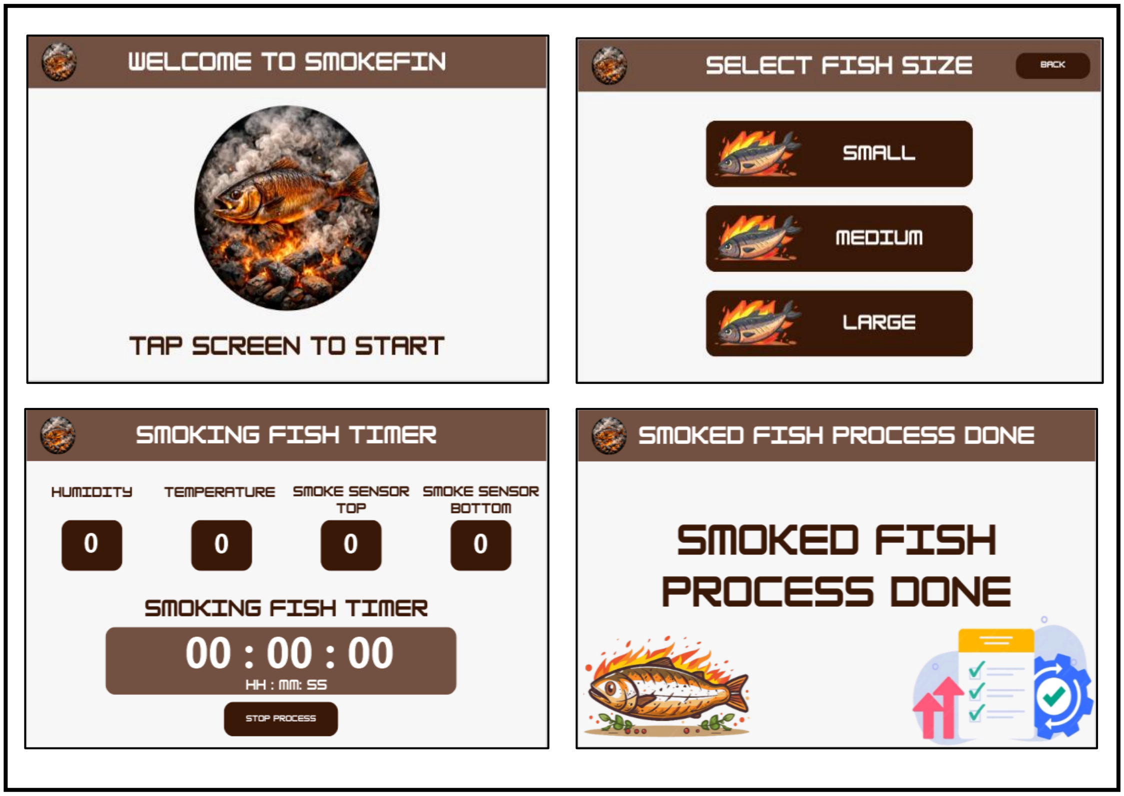


Figure 17. GUI of LCD screen of the SmokeFin Machine

Figure 17 shows the graphical user interface of the LCD screen of the SmokeFin machine. It starts with the home screen where the user starts the process. Next is the Fish Size Selection Screen that allows the user to select the size of the fish to be smoked, with options labeled Small, Medium, and Large. This selection is important because it determines the appropriate smoking duration. A Back button is provided to allow users to return to the previous screen if needed.

Once a fish size is selected, the system proceeds to the Smoking Fish Timer Screen, which displays real-time monitoring data and process status. This screen shows the current humidity and temperature inside the smoking chamber, ensuring optimal conditions for tinapa production. It also includes readings from the top and bottom smoke sensors, which help monitor smoke distribution and ensure uniform smoking throughout the chamber.

At the center of the screen is the Smoking Fish Timer, displayed in an hours-minutes-seconds (HH:MM: SS) format. This timer allows users to track the remaining or elapsed smoking time accurately. A Stop Process button is included to give users full control, enabling them to immediately halt the smoking operation if adjustments or safety actions are required. Once the process is done, the lcd displays "smoked fish process done".

Performance Evaluation

Performance evaluation is essential in determining how well the SmokeFin system meets its intended goals. It helps identify areas that need improvement, ensuring the machine delivers efficiency, safety, and user satisfaction. In this project, the evaluation focuses on several dimensions, including functionality, reliability, energy efficiency, food safety, wireless connectivity, and overall user experience.

Table 1. Likert Scale

Scale	Statistical Limit	Verbal Interpretation
4	3.50 – 4.0	Very Acceptable
3	2.50 – 3.49	Acceptable
2	1.50 - 2.49	Not Acceptable
1	1.00 – 1.49	Very Unacceptable

A Likert Scale will be applied to measure the quality and effectiveness of the system, providing valuable insights into its strengths and weaknesses. This approach will allow developers to refine both the hardware and software, ensuring that SmokeFin consistently produces high-quality Tinapa while remaining safe, sustainable, and user-friendly.

Ethical and Safety Considerations

Developing SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production requires strict ethical and safety considerations. As the system involves both food production and IoT-based automation, it must ensure fairness, sustainability, user safety, and consumer protection. The project is designed not only to improve efficiency and productivity but also to safeguard the health of both operators and consumers while promoting ethical technology use. The evaluation of the system is supported by internationally recognized standards such as ISO 14159:2002 – Safety of Machinery: Hygiene Requirements for the Design of Machinery, which ensures hygienic machine design suitable for food processing environments, and ISO 25010:2023 – Systems and Software Quality Requirements and Evaluation, which assesses the quality, reliability, safety, maintainability, and performance efficiency of the software and IoT components used in the system.

Ethical Considerations

1. **Fair Labor Practices.** The system must support rather than replace the livelihood of small-scale Tinapa producers. SmokeFin should complement workers by making the smoking process safer, more efficient, and less labor intensive, without causing job displacement or exploitation.
2. **Green and Sustainable Production.** Environmental sustainability is a key principle. The project aims to reduce smoke emissions, use renewable by-products (e.g., coconut shells) as fuel, and minimize energy waste. This ensures that fish smoking aligns with eco-friendly practices and reduces environmental impact.
3. **Data Privacy and Security.** Since SmokeFin incorporates IoT and real-time monitoring, any collected data (temperature, humidity, production logs) must be protected. Compliance

with ISO/IEC 27001:2022 ensures that user information and system data are secure from misuse or unauthorized access.

4. **Accessibility and Inclusivity.** The project should remain affordable and practical for small-scale vendors and cooperatives in coastal communities. It must avoid creating a technology gap by ensuring that even low resource users can access and benefit from the system.
5. **Product Safety and Consumer Rights.** The SmokeFin must comply with ISO 22000:2018 on Food Safety Management to ensure that smoked fish is hygienic, safe, and free from harmful contaminants.

Safety Considerations

1. **Machine Operation Safety.** The system must include clear safety features such as emergency shut-off, automatic regulation of temperature and humidity, and warning signals to minimize accidents during operation. A user manual with proper training instructions should also be provided.
2. **Electrical and Mechanical Safety.** As the system integrates microcontrollers, heating elements, and actuators, electrical components must be insulated and grounded. Regular maintenance and inspection must be enforced to avoid electrical shocks, overheating, or mechanical breakdowns.

3. **High Temperature and Smoke Exposure.** Since fish smoking involves heat and smoke, the system should be designed with protective insulation, proper ventilation, and controlled smoke flow. This prevents burns, scalds, and prolonged smoke exposure that could harm operators' health.
4. **Food Safety and Contamination Prevention.** All parts in direct contact with fish must be made of food-grade, nontoxic, and easy-to-clean materials. Proper sealing and sanitation procedures should be followed to prevent microbial or chemical contamination during the smoking process.
5. **Ergonomic and User-Friendly Design.** The system should be designed with operator ergonomics in mind. Controls must be simple, accessible, and easy to operate. The layout should minimize repetitive strain or unsafe manual handling, making the machine safe and efficient to use.

Project Cost Analysis

Cost analysis determines whether the proposed system is financially practical and sustainable in the long run. It looks into everything needed to bring the project to life—from the cost of materials and equipment to labor and other resources. The list below shows the materials and components that need to be purchased to make the system possible.

Table 2. Cost Analysis of the Project

MATERIALS	SPECIFICATION	PRICE	QTY	SUBTOTAL
Arduino Mega		1,265	1	1,265
Arduino Mega Terminal Board		413	1	413
Esp32 with Expansion Board (Type C)		515	1	515
7 Inches Nextion LCD		5277	1	5277
Electric Stove	1000 W	190	2	380
Solid State Relay 40A	40 DA	179	2	358
Lm2596 Buck Converter		49	2	98
9x15 Universal PCB Green		65	1	65
H Bridge Motor Driver	BTS7960 43A	339	1	339
12v Linear Actuator		3850	1	3850
2 Pin Terminal Blocks		109	1	109
3 Pin Terminal Blocks		165	1	165
MQ135 Smoke Sensor		109	2	218
DS3231 RTC Module		140	1	140
DHT22 Temperature Sensor		220	1	220
Stranded Wires Black		165	2	330
Stranded Wires Red		165	1	165
Stranded Wires 15 Meters		165	1	165
Jumper Wires Male to Female		75	2	150
Jumper Wires Male to Male		75	1	75
Jumper Wires Female to Female		75	1	75
High Pressure Water Pump	12V	321	1	321
Relay Module		105	1	105

MATERIALS	SPECIFICATION	PRICE	QTY	SUBTOTAL
Switching Power Supply	12v 10a	308	1	308
Plug	220	49	1	49
Shrinkable Tube	4mm	19	1	19
Shrinkable Tube	5mm	17	5	85
Heat Resistant Wire 3 meters		97	3	291
Single Phase outgoing terminal		170	1	170
Car Speaker		212	1	212
Memory Card		188	1	188
Core Male to female Audio		150	1	150
DFPlayer Mini MP3 Player Module		50	1	50
PAM8610 Digital Power Amplifier		100	1	100
PAM8610 Digital Power Amplifier		100	1	100
Board Protective Glass		3000	1	3000
			Subtotal	19,520
Casing				50,000
Documentation				5,000
			Total	74,520

Chapter IV

RESULTS AND DISCUSSION

This chapter presents the results of testing and evaluating the SmokeFin based on the study criteria. The machine was assessed in terms of safety, functionality, efficiency, usability, and durability, while the mobile application was evaluated for functionality and usability. The findings were used to determine the effectiveness of the system in improving traditional tinapa production through automation and IoT integration.

Presentation of Results

The SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production was successfully developed to improve the efficiency, safety, and consistency of the traditional fish smoking process. By integrating sensors for temperature, humidity, and smoke level detection, the system was able to provide real-time monitoring of smoking conditions throughout the process.

The mobile monitoring application, enabled users to remotely observe the smoking operation and respond immediately to emergency situations through the emergency stop feature. This innovation improved operational safety by reducing user exposure to heat and smoke while ensuring more consistent smoking results. Furthermore, the system enhanced the quality of tinapa production by monitoring a smoking environment, resulting in a more uniform and hygienic smoked fish output. The following are the results and discussion of the project's functionality. The SmokeFin system successfully enhanced the fish smoking process by automating essential operations and providing real-time monitoring capabilities, resulting in improved production efficiency, safer operation, and more reliable smoked fish quality.

1. Development of the SmokeFin Machine

As the SmokeFin system consists of both a physical fish smoking machine and a mobile monitoring application, the design and development phase involved the creation of the machine's structural layout and electronic integration plans. The initial physical design of the SmokeFin machine was developed using AutoCAD, where the overall dimensions, smoking chamber, fish racks, smoke box, and control compartments were modeled. Based on the selected hardware components, block diagrams and wiring diagrams were then drafted to define the interaction between the Arduino Mega 2560, ESP32 module, sensors, relays, heating elements, and actuators. At the same time, the logical operation of the machine and the mobile application was illustrated through system flowcharts to establish the sequence of processes from fish size selection to automated smoking and emergency response.

Figure 18. Front View of the Smokefin Machine

Figure 18 shows the front right view of the SmokeFin machine, where the main external parts of the system can be clearly seen. The machine was first designed using 3D modeling software to properly visualize the structure, dimensions, and placement of each component before fabrication. The development of the machine started by creating the main frame using durable metal materials to ensure stability and support for the entire structure. After the frame was completed, the outer panels and glass doors were installed to provide protection and allow easy monitoring of the smoking process inside the machine. The racks were then positioned inside the chamber to hold the fish products during operation. the smoke box was installed at the lower portion of the machine, where the wood dust or smoking material is placed to generate smoke. The cable conduit was also added to organize and protect the electrical wires connected to the different components of the system. Afterward, the Nextion LCD display was mounted on the upper front section of the machine to allow easy monitoring and control of the system parameters.

Figure 19. The Smoke Box of Smokefin Machine

Figure 19 shows the smoke box and its internal components of the SmokeFin machine. The smoke box was developed to serve as the main chamber where smoke is generated using wood dust during the smoking process. The fabrication started by constructing a durable metal enclosure capable of withstanding high temperatures produced during operation. Inside the smoke box, heating elements were installed at the bottom section to produce heat needed for burning the wood dust and generating smoke. These heating elements were carefully positioned and secured to ensure stable heat distribution and efficient smoke production. Around the heating elements, protective insulation materials were added to minimize heat loss and improve safety during operation. A sprinkler system was also installed inside the smoke box as a safety feature to control excessive smoke or fire. The sprinkler was connected to the water supply system and was designed to release water whenever needed to stop continuous smoke production or prevent overheating inside the chamber. Proper piping and fittings were attached to ensure smooth water flow toward the sprinkler system. In addition, an ash box was placed at the lower portion of the smoke box to collect the ashes produced after the wood dust was burned. The ash box was designed to be removable for easier cleaning and maintenance after operation. All internal components were arranged carefully to ensure safe smoke generation, efficient operation, and convenient maintenance of the SmokeFin machine.

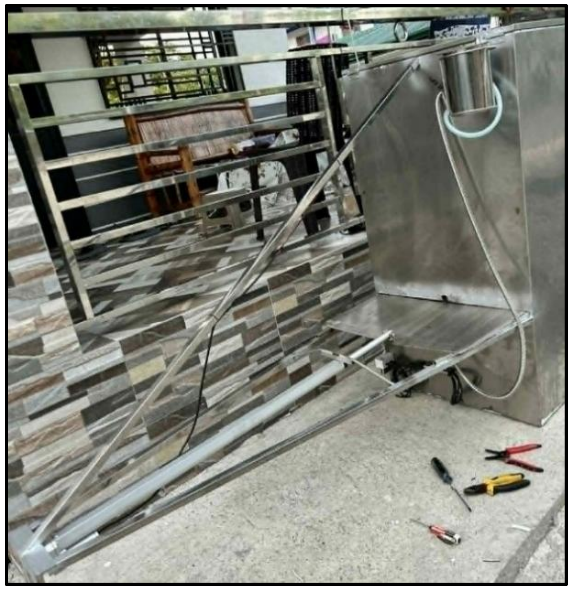


Figure 20. Left-Rear View of the SmokeFin Machine

Figure 20 shows the left rear view of the SmokeFin machine, where the water tank, hose connections, motor, and actuator can be seen. The rear section of the machine was developed to accommodate the supporting components responsible for smoke control and machine operation.

The fabrication process started with the installation of the steel water tank at the rear upper section of the machine. The tank was designed using durable metal materials to safely store water needed for the sprinkler system. After mounting the water tank, hoses were connected from the tank going to the smoke box to provide a pathway for water flow toward the sprinkler system during operation.

Next, the motor and actuator were installed near the smoke box mechanism. The actuator was designed to automatically open and close the smoke box passage to control the flow of smoke entering the fish chamber. This mechanism helps prevent excessive smoke from entering the layers of fish and allows better control of the smoking process. The motor was securely mounted and connected to the actuator to ensure smooth and stable movement during operation.

Figure 21. Wiring Box of the SmokeFin Machine

Figure 21 shows the internal wiring connections of the SmokeFin machine. After developing the machine's casing and integrating its major features, the developers proceeded with the wiring and electrical connections of the system. Electrical components such as the sensors, power supply, relays, and microcontroller were carefully interconnected to ensure proper communication and functionality of the automated features. The wiring arrangement was organized systematically to promote safety, reliability, and efficient operation of the machine. Proper cable management and insulation were also applied to minimize the risk of short circuits and electrical hazards during operation. This stage of development was essential in ensuring that the SmokeFin system could perform its monitoring and automation processes accurately and consistently.

2. Development of the User Interface of Smokefin

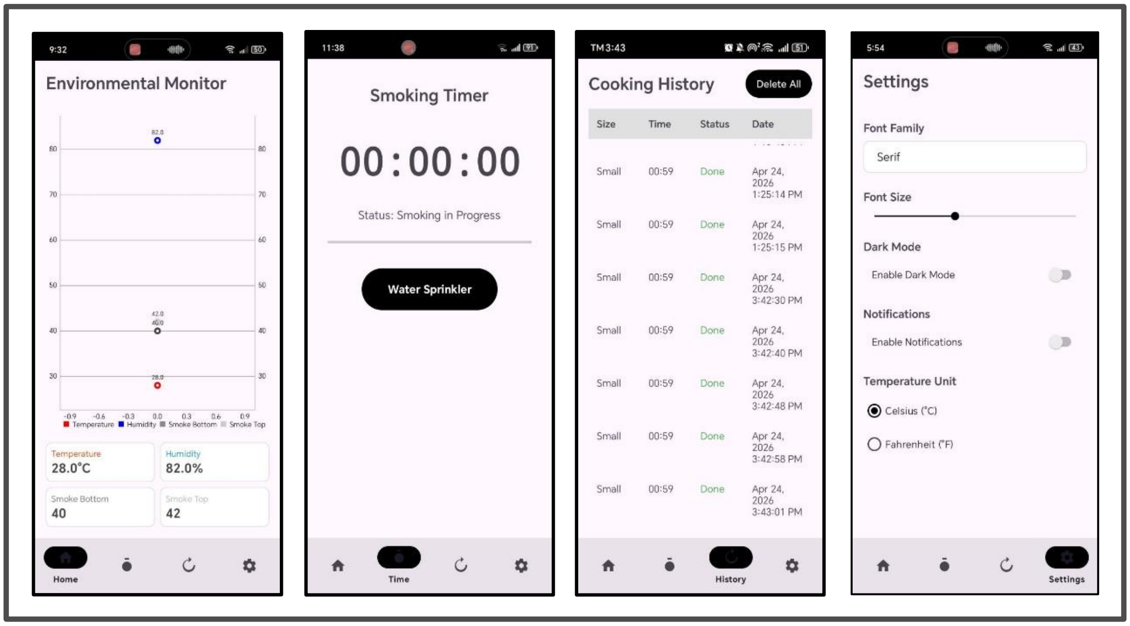


Figure 22. Mobile Application of SmokeFin

Figure 22 shows the mobile application monitoring system developed for the SmokeFin machine. The application was designed to provide remote monitoring and easier access to the machine's operational status through a smartphone device. The development process started by planning the interface and features needed in the application, including temperature monitoring, smoke status, notifications, and system updates. After planning the design, the application was connected to the machine's control system through wireless communication technology to allow real-time data transmission between the SmokeFin machine and the mobile device. The necessary programming and integration were performed to ensure that the application could properly receive and display system information from the machine. User interface elements such as buttons, status indicators, and monitoring panels were arranged clearly to make the application simple and easy to use. The system was tested multiple times to verify connectivity, responsiveness, and accuracy of the displayed data. Necessary improvements were made to ensure reliable and convenient remote monitoring of the SmokeFin machine.

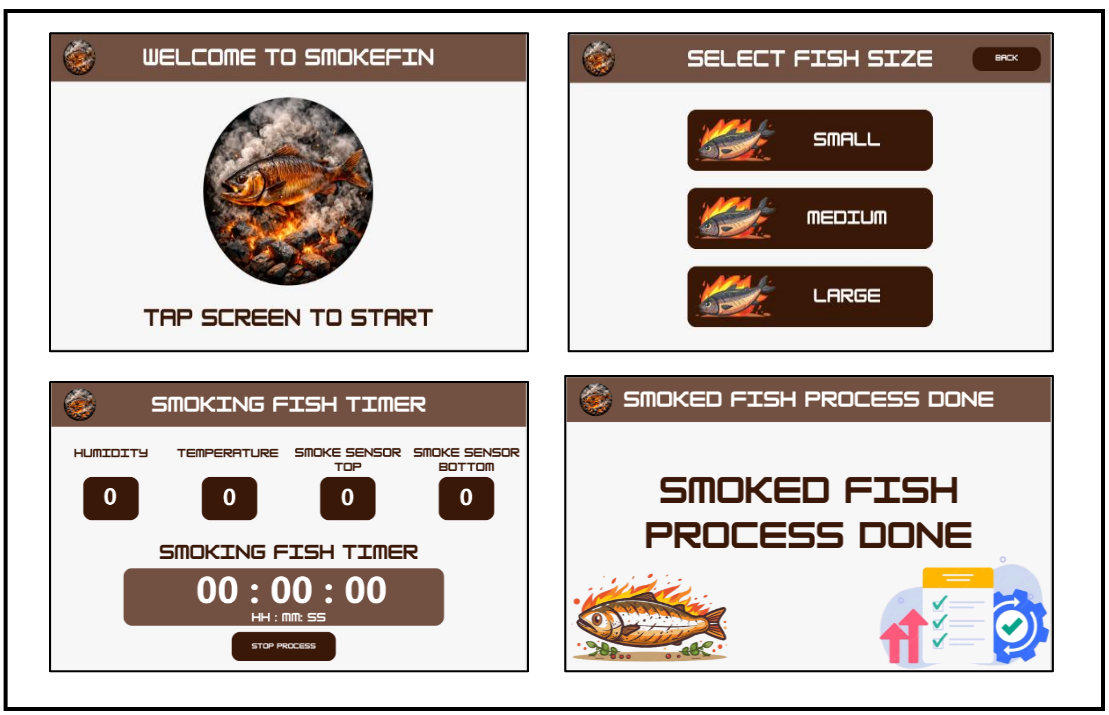


Figure 23. LCD Screen of Smokefin

Figure 23 shows the Nextion LCD display interface used in the SmokeFin machine. The LCD interface was developed to provide a user-friendly monitoring and control system for the machine during operation. The development process started by designing the display layout using the Nextion Editor software, where the necessary information such as temperature readings, system status, timer, and control buttons was arranged properly for easier viewing and operation. After completing the interface design, the Nextion LCD display was connected to the microcontroller of the SmokeFin machine to allow communication between the hardware and the display system. The necessary programming and configurations were then performed to ensure that real-time data from the sensors and other components could be displayed correctly on the screen. The LCD was mounted on the front panel of the machine for convenient access and monitoring by the user. Several tests and adjustments were conducted to improve the responsiveness, readability, and overall performance of the interface, making the system more efficient and user-friendly during operation.

Smoked-fish Production Metrics

Efficient fish smoking production requires systematic monitoring of key operational parameters to ensure product quality, process efficiency, and sustainability. This section evaluates two critical performance metrics of the SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production. Smoking duration, and smoked fish quality output. These quantitative measurements allow operators to assess the machine's effectiveness, optimize smoking conditions, and maintain consistent product standards. The collected data serves as a basis for performance evaluation and continuous improvement of the SmokeFin system in producing high-quality and sustainable tinapa. In addition, monitoring these metrics helps identify possible inconsistencies during the smoking process that may affect the final product. Accurate evaluation of production performance also supports better resource management, energy efficiency, and reduced material waste. Through continuous monitoring and analysis, the SmokeFin system can contribute to a more reliable, safe, and technologically enhanced fish smoking process. Overall, the system promotes greater efficiency, product quality, and sustainability in modern tinapa production.

Table 3 presents a comparative analysis of the Smoking Duration of Fish, Input and Output of the Traditional method and the SmokeFin system. The result shows that the SmokeFin system improved both the smoking duration and the production output compared to the traditional method of smoking process. In the traditional method, smoking fish typically takes 3 to 5 hours, while smokefin system reduces time to 1 to 3 hours. In terms of product input, traditional method uses mixed sizes of fish while smokefin system classifies fish into small, medium, large categories for a more organized smoking process. For the product output, the traditional method process produces around 5 to 7 kilograms of smoked fish per batch, while the SmokeFin system increases the yield to 8 to 14 kilograms. Overall, the comparison demonstrates that SmokeFin enhances productivity by reducing smoking time and increasing production capacity, making it a more efficient and sustainable solution for tinapa production.

Table 3. Comparative Analysis of Smoking Fish Duration, Process Input and Output of Traditional and SmokeFin System

Process	Traditional	SmokeFin
Smoking Duration	3-5 hours	Small (1 hour) Medium (2 hours) Large (3 hours)
Product Input	Mixed	Small (4-5 inches) Medium (6-8 inches) Large (8-12 inches)
Product Output	5-7 kgs	8-14 kgs

Evaluation of the Project

The following are the results of the project's evaluation by the beneficiaries based on its hygienic design, system functionality, usability, efficiency, operational safety, durability, mobile application functionality, mobile application usability and the overall evaluation.

Table 4 assesses the sanitary standards and construction of the smoking machine across five specific criteria. The project garnered an average weighted mean of 3.74, resulting in a verbal interpretation of "Very Acceptable". The highest-rated feature was the hygiene of the fish racks, which earned a perfect score of 4.0, while the protective smoke box design followed closely at 3.9. Even the lowest-ranked item—ease of cleaning the internal chamber—maintained a high score of 3.5. These figures demonstrate that the machine's physical build effectively prioritizes food safety and contamination prevention.

Table 4. Hygienic Design of the Physical Machine

Hygienic Design of the Physical Machine	Weighted Mean	Rank	Verbal Interpretation
1.1 The machine is made of safe, food-grade materials that will not contaminate the fish.	3.7	3	Very Acceptable
1.2 The inside of the chamber is easy to clean after each use.	3.5	5	Very Acceptable
1.3 The smoke box keeps ash and dirt away from the fish.	3.9	2	Very Acceptable
1.4 The design of the machine prevents pests or dust from getting inside.	3.6	4	Very Acceptable
1.5 The fish racks keep and hold the product hygienic.	4	1	Very Acceptable
Average	3.74	-	Very Acceptable

Table 5 evaluates the technical performance and reliability of the machine's core components. The system achieved an overall average weighted mean of 3.55, which translates to a "Very Acceptable" rating. The linear actuator's ability to properly close the smoke box was the top-performing feature with a score of 3.9, while sensor accuracy for heat and smoke followed at 3.8. Although the consistency of the smoking process received a lower score of 3.0, it remained within the "Acceptable" range. The results suggest that the SmokeFin system is capable of providing stable and reliable operation during the fish smoking process. Furthermore, the evaluation confirms that the integration of IoT-enabled components contributes positively to the overall functionality and performance of the system.

Table 5. System Functionality

System Functionality	Weighted Mean	Rank	Verbal Interpretation
2.1 The system successfully performs the fish smoking process.	3.5	3	Very Acceptable
2.2 The sensors accurately read the heat and smoke levels	3.8	2	Very Acceptable
2.3 The linear actuator closes the smoke box properly when needed.	3.9	1	Very Acceptable
2.4 The system operates consistently during the smoking process.	3	4	Acceptable
Average	3.55	-	Very Acceptable

Table 6 measures the usability and ease of operation for the physical machine. With an average weighted mean of 3.68, the machine's usability is classified as "Very Acceptable". The clarity and ease of use of the LCD screen led the rankings with a score of 3.9, followed by the ease of arranging fish racks at 3.8. Both the manual wood dust loading and general ease of learning the operation scored 3.6, showing consistent user satisfaction. These results suggest that the machine is intuitive for operators to manage without complex training. Overall, the result indicate that the SmokeFin system provides a user-friendly interface and straightforward operation suitable for practical use.

Table 6. Usability

Usability	**Weighted Mean**	**Rank**	**Verbal Interpretation**
3.1 The LCD screen of the machine is easy to understand and use.	3.9	1	Very Acceptable
3.2 The e-data such as temperature, smoke level, and time on the LCD screen are easy to read.	3.5	5	Very Acceptable
3.3 The wood dust is easy to put into the machine manually.	3.6	3.5	Very Acceptable
3.4 Fish racks are easy to arrange inside the machine.	3.8	2	Acceptable
3.5 The overall operation of the machine is easy to learn and use.	3.6	3.5	Very Acceptable
Average	3.68	-	Very Acceptable

Table 7 examines the productivity and quality output of the machine compared to traditional methods. The project received an average weighted mean of 3.13, earning a verbal interpretation of "Acceptable". The quality of the produced "Tinapa" was the highest-rated aspect at 3.4, followed by the system's ability to improve productivity at 3.2. While the duration of the smoking process and the even distribution of heat received lower scores of 2.9 and 3.0 respectively, they still met the required standards. This section indicates that while the machine is effective, there is room for optimization in heat distribution and timing.

Table 7: Efficiency

Efficiency	**Weighted Mean**	**Rank**	**Verbal Interpretation**
4.1 The smoke and heat spread evenly across all fish racks.	3.0	3	Acceptable
4.2 The smoking duration is efficient and reasonable.	2.9	4	Acceptable
4.3 The system improves productivity compared to traditional methods.	3.2	2	Acceptable
4.4 The quality of the smoked fish (Tinapa) is very satisfactory.	3.4	1	Acceptable
Average	3.13	-	Acceptable

Table 8 focuses on the safety features and protective measures integrated into the machine. The evaluation resulted in an average weighted mean of 3.18, falling under the "Acceptable" verbal interpretation. Accessibility to emergency stops and breakers was the highest-rated safety feature at 3.7, while the enclosure of electrical components followed at 3.5. The lowest score of 2.5 was given to the volume of sound alerts, suggesting a need for louder alarms. Generally, the results confirm that the machine is safe to handle, though auditory warnings could be enhanced.

Table 8: Operational Safety

Operational Safety	**Weighted Mean**	**Rank**	**Verbal Interpretation**
5.1 Electrical and heating components are properly enclosed.	3.5	2	Very Acceptable
5.2 Safety features such as breakers, and emergency stop buttons are easy to reach.	3.7	1	Very Acceptable
5.3 All the wiring is safe and protected from water or heat.	3	4	Acceptable
5.4 Sound alerts are loud enough to hear when there is a problem.	2.5	5	Acceptable
5.5 The heat inside the machine does not burn the outer casing. The outer casing is safe to touch.	3.2	3	Acceptable
Average	3.18	-	Acceptable

Table 9 evaluates the technical performance and reliability of the project's companion mobile app. The application achieved a "Very Acceptable" rating with an average weighted mean of 3.6. The accuracy of real-time data display (temperature, smoke, and time) was the top feature at 3.9, while the reliability of labeled buttons followed at 3.8. While app responsiveness during data refreshes was the lowest-ranked at 3.2, the overall connectivity remained strong at 3.6. These scores indicate that the digital monitoring system is a reliable tool for remote oversight.

Table 9. Mobile Application Functionality

Mobile Application Functionality	Weighted Mean	Rank	Verbal Interpretation
6.1 The mobile app accurately displays real-time data such as temperature, smoke level, and smoking duration from the machine.	3.9	1	Very Acceptable
6.2 The app responds quickly and reliably when refreshing or updating monitoring data	3.2	5	Acceptable
6.3 The app consistently connects to the system without errors or disconnections.	3.6	3	Very Acceptable
6.4 All features and buttons are working according to their label.	3.8	2	Very Acceptable
6.5 The app sends a clear notification when the fish is done	3.5	4	Very Acceptable
Average	3.6	-	Very Acceptable

Table 10 assesses how intuitive and well-designed the mobile application interface is for the user. The app earned a "Very Acceptable" interpretation with an average weighted mean of 3.56. Users found the information displayed very easy to understand (3.9) and the interface design clear and organized (3.8). Navigation for first-time users and the overall layout also scored well at 3.5 and 3.6 respectively.

Despite a slightly lower score for input response speed (3.0), the results show the app is user-friendly and well-structured.

Table 10: Mobile Application Usability

Usability	Weighted Mean	Rank	Verbal Interpretation
7.1 The mobile app is easy to navigate, even for first-time users.	3.5	4	Very Acceptable
7.2 The interface design is clear, organized, and user-friendly.	3.8	2	Acceptable
7.3 The information displayed on the application is easy to understand.	3.9	1	Very Acceptable
7.4 The app responds quickly to user inputs such as opening screens, and refreshing data.	3	5	Acceptable
7.5 The app layout helps users easily find important features and data.	3.6	3	Very Acceptable
Average	3.56		Very Acceptable

Table 11 provides a comprehensive overview of all evaluation categories to determine the project's total impact. The entire system received an overall average weighted mean of 3.49, resulting in a final verbal interpretation of "Acceptable". Hygienic Design was the highest-ranked category (3.74), followed by Usability (3.68) and Mobile Application Functionality (3.6). Efficiency and Operational Safety were the lowest-ranked categories, yet they remained within the acceptable performance threshold. This summary confirms that the project is a successful integration of hardware and software that meets the needs of its beneficiaries.

Table 11: Result Summary of the Evaluation

Overall Evaluation	Weighted Mean	Rank	Verbal Interpretation
Hygienic Design of the Machine	3.74	1	Very Acceptable
System Functionality	3.55	5	Acceptable
Usability	3.68	2	Very Acceptable
Efficiency	3.13	7	Acceptable
Operational Safety	3.18	6	Acceptable
Mobile Application Functionality	3.6	3	Very Acceptable
Mobile Application Usability	3.56	4	Very Acceptable
Average	3.49	-	Acceptable

Chapter V

CONCLUSIONS AND RECOMMENDATIONS

This chapter contains the conclusions that the developers observed throughout the project's development. Recommendations were also presented for future developers conducting similar design projects.

Conclusions

The SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production was successfully designed, developed, and evaluated based on the objectives of the study. From the results obtained, the researchers arrived at the following conclusions:

1. The SmokeFin system successfully improved the traditional fish smoking process by integrating IoT technology, enabling real-time monitoring of temperature, humidity, and smoke level, which resulted in more consistent and controlled smoking conditions. The developed system effectively reduced smoking time from 3–5 hours to 1–3 hours while increasing production output from 5–7 kilograms to 8–14 kilograms, demonstrating improved efficiency and productivity compared to the manual method.

2. The mobile application provided a convenient way to monitor the system remotely, improving accessibility and user experience, although control features were limited to selected functions.

3. The integration of hardware components such as sensors, microcontrollers, and actuators enabled semi-automation of the smoking process, reducing manual labor and minimizing exposure of users to heat and smoke.

4. Based on the evaluation results, the SmokeFin system was rated as Very Acceptable in terms of hygienic design, functionality, usability while rated as Acceptable in terms of efficiency, and operational safety. Mobile application functionality and usability are also rated as Very Acceptable and overall evaluation of the system is rated as Acceptable, indicating that the system meets the required standards for small-scale tinapa production.

Recommendations

To further recommendations are proposed. improve of the system, several

1. Future developers may enhance the system by implementing automatic control of temperature, humidity, and smoke level to achieve a fully automated smoking process.

2. A fish readiness detection system may be developed using sensors or image processing to determine if the fish is already properly smoked, reducing reliance on manual observation.
3. The system may be improved by adding offline functionality or local network connectivity to ensure continuous operation even without internet access.
4. The mobile application may be enhanced by adding full remote-control features, notifications, and alert systems for abnormal conditions such as overheating or system failure.
5. Future studies may focus on scaling the system for commercial use, allowing it to handle larger production volumes while maintaining efficiency and consistency.
6. The integration of alternative energy sources such as solar power is recommended to improve sustainability and reduce operational costs.
7. Additional safety features, such as automatic shutdown mechanisms and improved fire detection systems, may be implemented to further enhance user safety.
8. Future researchers may explore the use of advanced materials and improved structural design to enhance durability, hygiene, and overall system performance.

BIBLIOGRAPHY

[1] R. M. Davies and O. A. Davies, "Traditional and Improved Fish Processing Technologies in Bayelsa State, Nigeria," *European Journal of Scientific Research*, vol. 26, no. 4, pp. 539-548, 2009.

[2] O. M. Osineye *et al.*, "Comparative Evaluation of Performances of Biomass-Sourced Briquette (Sawdust), Wood and Charcoal as Fuelling Materials in Fish Smoking," *Journal of Sustainable Development*, vol. 17, no. 1 & 2, pp. 106-111, Sep. 2020.

[3] International Organization for Standardization. (2002). *ISO 14159:2002 Safety of machinery—Hygiene requirements for the design of machinery*. Available Online: https://www.iso.org/standard/23150.html

[4] Suryadi, D. A., & Sulistiyani, E. (2022). Evaluation of information quality using ISO/IEC 25010:2011 (Case research: Menu Harianku application). International Journal of Innovation in Enterprise System, 6(2), 95-102. Available Online: https://doi.org/10.25124/ijies.v6i02.167

[5] A. M. Djordjevic, "Smoke and Smoked Fish Production," *Meat Technology*, vol. 61, no. 2, pp. 93-102, 2020.

[6] E. Nurmianto, A. Anzip, and Sampurno, "Design of mobile, portable and ergonomic fish smoke machine on a motorcycle," *IOP Conference Series: Materials Science and Engineering*, vol. 674, no. 1, p. 012047, Nov. 2019

[7] M. I. Joesidawati, Suwarsih, A. W. Nuruddin, and Sriwulan, "Performance Test Smoked Tool Fish Which Effective, Hygienic, and Eco-Friendly," *Revista de Gestão Social e Ambiental*, vol. 18, no. 6, p. e07478, Jun. 2024, doi: 10.24857/rgsa. v18n6-119.

[8] E. Fradinata and Z. M. Kesuma, "Design of Fishing Smoke (Saltization) Equipment and Its Process for Improving the Economy Community in Lampulo Banda Aceh," *J. Pengabdian Pada Masyarakat*, vol. 5, no. 1, pp. 213-220, Feb. 2020. doi:10.30653/002.202051.250

[9] Rasco, B. Smoking Fish at Home Safely, PNW238. Pacific

Northwest Extension publications—Washington State University Extension. 2009. Available online: http://www.uaf.edu

[10] Tsai, W.T.; Lee, M.K.; Chang, Y.M. Fast pyrolysis of rice straw, sugarcane bagasse and coconut shell in an induction-heating reactor. *J. Anal. Appl. Pyrolysis* 2006, *76*, 230–237.

[11] A. J. R. Taduran, "The Development of a Microcontroller based Smoked Fish Machine," *International Journal of Engineering Trends and Technology*, vol. 72, no. 3, pp. 370–380, Mar. 2024, doi: 10.14445/22315381/IJETTV72I3P132.

[12] Mendoza, Xavier & Abug, Nerisa. (2025). Tinapa Production: An Analysis of Current Practices, Challenges, and Potential for Enhancement in the Province of Cavite. 1-13. 10.17613/a2ryj-wrx87.

[13] Pramanik, Sabyasachi, Soma Bag, Atanu Roy, and Pranati Rakshit. 2025. Comprehensive Modernity of Fish Smoking Techniques Using the IoT., 165–190. doi:10.4018/979-83373-4332-7.ch006.

[14} R. J. Setiawan, K. Ma'ruf, D. Darmono, N. Azizah, and N. E. Khosyiati, "Systematic modernization of fish smoking method with the implementation of smoked fish machine based on Internet of Things technology," *Int. J. Systematic Innovation*, vol. 9, no. 1, pp. 57–67, 2025, doi: 10.6977/IJoSI.202502_9(1).0005.

APPENDIX A

Design Project Assignment Form

Project Title: SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production

Name and signature	Project Role	Email Address/ Contact Number
Ruzzel A. Livelo	Main Researcher and Software Developer	ruzzellivelo911@gmail.com 09653469274
Jayperson M. Caya	Main Researcher and Hardware Developer	jaypersonmarasigacaya@gmail.com 9660296909
Yasme O. Fajarito	Main Researcher/ Technical Writer	yasmefajarito@gmail.com 09278070832
John Aeron P. Fruelda	Main Researcher/Project Designer	johnaeronfruelda@gmail.com 09657746920
Denmark G. Acosta	Assistant Technical Writer	denmarkgalacanacosta@gmail.com 09530457012

APPENDIX B

Design Project Topic Proposal Form

CPE DESIGN PROJECT TOPIC PROPOSAL	
Proposed Title	**Smart Smoke:** An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production
Name of Student/ Course, Year & Section	Denmark G. Acosta Jayperson M. Caya Yasme O.Fajarito John Aeron P. Fruelda Ruzzel A. Livelo BSCpE 3-1
Introduction	In many coastal and riverside communities, fish is more than just a daily meal—it's a way of life. But fresh fish, as valuable as it is, doesn't stay fresh for long. That's where tinapa, or smoked fish, comes in—not just as a delicious Filipino staple, but as a smart, time-honored way of preserving food. For generations, families have turned to smoking as a means to keep their catch from spoiling, especially when refrigeration was out of reach. The function of smoke tinapa goes beyond preservation. It enhances flavor, adds value, and helps small-scale fishers and vendors earn more from every catch. It gives them the power to store, transport, and sell fish without racing against time. Smoking fish transforms a perishable product into something lasting, marketable, and deeply rooted in tradition. In every piece of tinapa lies a blend of culture,

survival, and resilience—preserving not just fish, but livelihoods and legacies [1].

For generations, families have relied on fish not only for their own meals but as a vital source of income. Yet one challenge remains the same: how to preserve the day's catch before it spoils, especially when access to refrigeration is limited.

Traditionally, fish smoking is done over open fires, requiring hours of constant attention, physical effort, and exposure to smoke and heat. While the method has stood the test of time, it can be inefficient, inconsistent, and even hazardous to health. That's where the smoke tinapa machine comes in—not as a replacement for tradition, but as an improvement.

The machine is built to make the process faster, safer, and more consistent. It helps produce higher-quality smoked fish that lasts longer, tastes better, and meets market standards. Most importantly, it lightens the burden on those—often women and elders—who do the work by hand. For small-scale fisherfolk and vendors, the smoke tinapa machine is more than just a tool; it's an opportunity to increase earnings, reduce waste, and continue a meaningful tradition with dignity and efficiency.

By blending technology with tradition, this machine honors the past while opening doors to a better future for the communities who rely on tinapa not just for food, but for life [2]. Despite being widely used, traditional fish smoking techniques frequently lack efficiency, consistency, and environmental control, which results in inconsistent product quality and restricted traceability. By creating an Internet of Things (IoT)-enabled tinapa maker that automates and tracks the smoking process, this project seeks to address these issues. The device gives real-time data on important aspects, including temperature, humidity, and smoking duration, by combining wireless technology with sensors. This innovation aims to improve the quality and dependability of smoked fish production, especially for community-based cooperatives, small-scale fish processors, and local business owners. In addition to enhancing product uniformity, the system's more regulated and data-driven methodology promotes improved food safety procedures, energy efficiency, and scalability for possible commercial application. With the help of this IoT solution, users may modernize established methods and make them more traceable, sustainable, and user-friendly. This project aligns with several Sustainable Development Goals (SDGs), particularly SDG 2 (Zero Hunger), by supporting food security through improved post-harvest practices; SDG 8 (Decent Work and Economic Growth), by

	reducing labor intensity and boosting productivity for farmers; and SDG 9 (Industry, Innovation, and Infrastructure), by promoting the adoption of smart agricultural technologies and modernizing rural farming systems.
Objectives of the Study	The main objective of this project is to develop Smart Smoke: an Internet of Things (IoT) tinapa maker that automates and monitors the fish smoking process to improve efficiency, ensure consistent quality, and provide real-time data on temperature, humidity, and smoking duration for better control and traceability. Specifically, the following are the specific objectives of this project: 1. To design an IoT-enabled fish smoking system that tracks key conditions like temperature, humidity, and time to ensure consistent results. 2. To develop a microcontroller-based platform that automates the smoking process and can be easily controlled via a mobile or web app. 3. To integrate essential hardware sensors, heating elements, and wireless modules into a compact and efficient prototype for small-scale use. 4. To test and evaluate the system's performance in terms of reliability, efficiency, and product quality, while

	identifying potential challenges like connectivity, food safety, and scalability.
Scope and Limitation of the Study	This study focuses on designing and developing an IoT-based tinapa (smoked fish) maker that aims to modernize the traditional fish smoking process. It covers the creation of a smart system that uses sensors to monitor temperature, humidity, and smoking time and integrates a microcontroller that allows users to automate and control the process remotely through an app or website. The goal is to improve efficiency, ensure consistent quality, and provide real-time monitoring during the smoking process. However, the project is limited to small-scale production and may not fully reflect the complexities of commercial operations. It also assumes stable internet connectivity, which may not always be available in remote fishing communities. Additionally, while the system focuses on environmental factors like heat and humidity, it does not cover other food safety standards such as chemical contamination. Finally, the economic feasibility and long-term impact of using this technology on a wider scale are beyond the scope of this study.
Methodology	Agile process model, which usually lasts between one and four weeks; and the division of the entire project into smaller parts helps to reduce project risk and the overall project

delivery time requirements. Each iteration involves a team working through a fill software development life cycle including planning, requirements analysis, design, coding, and testing before a working product is demonstrated to the client.

Figure 1. Agile Methodology

Plan

The researchers began by identifying the primary objectives of the project: to automate the traditional fish smoking process, monitor critical factors such as temperature and humidity, and allow users to control the system remotely via a web or mobile interface. The needs of small-scale fish vendors and community users were also considered during this phase.

Design

In this stage, the researchers developed a system layout that combined both hardware and software elements. They selected appropriate components such as sensors, microcontrollers, and heating elements, and designed a safe and efficient smoking chamber. A simple and intuitive user interface for mobile or web access was also planned.

Develop

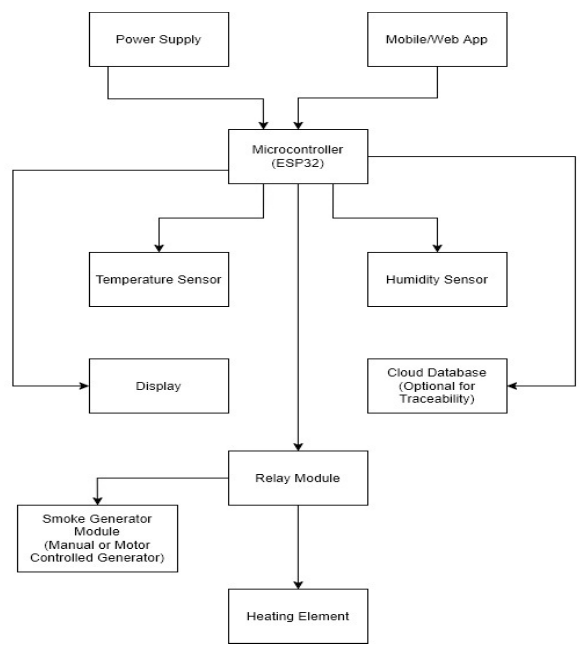
The researchers proceeded to build the prototype. This included programming the microcontroller to process sensor data and control the heating system. At the same time, a user-friendly application was developed to display real-time environmental data and provide remote access to the smoker's functions.

Test

The system was thoroughly tested to verify its accuracy, consistency, and reliability. The researchers ensured that the machine could maintain proper smoking conditions and respond correctly to user commands. They also evaluated the quality of the smoked fish and compared the results to those of traditional methods.

Deploy

Once the system had passed functional testing, the researchers integrated all components and finalized the prototype for

	actual use. The focus during deployment was on creating a fully operational, safe, and user-friendly device suitable for small-scale fish smoking. **Review** In the final phase, the researchers collected feedback from test runs, documented the outcomes, and identified areas for improvement. Limitations such as the dependency on internet access and the exclusion of advanced food safety testing were acknowledged, and suggestions were made for future development and enhancement of the system.
Proposed Design	 **Figure 2.** Block Diagram of the Proposed System The Mobile/Web App allows users to remotely monitor and control the smoker, while the ESP32 microcontroller acts as the central

	brain that reads data from the temperature and humidity sensors, manages output to the display for local status updates, and communicates with a cloud database for optional remote logging and traceability. Based on the sensor data, the ESP32 controls a relay module, which safely switches power to the heating element that regulates internal temperature, and optionally to a smoke generator module, which may be manually operated or motor-controlled to deliver consistent smoke.
References	[1] R. M. Davies and O. A. Davies, "Traditional and Improved Fish Processing Technologies in Bayelsa State, Nigeria," *European Journal of Scientific Research*, vol. 26, no. 4, pp. 539-548, 2009. [2] O. M. Osineye *et al.*, "Comparative Evaluation of Performances of Biomass-Sourced Briquette (Sawdust), Wood and Charcoal as Fuelling Materials in Fish Smoking," *Journal of Sustainable Development*, vol. 17, no. 1 & 2, pp. 106-111, Sep. 2020. [3] A. J. R. Taduran, "Comprehensive Modernity of Fish Smoking Techniques Using the IoT," *arXiv preprint arXiv:2404.13051.*

Prepared by:

Acosta, Denmark G.
BSCpE 3-1

Caya, Jayperson M.
BSCpE 3-1

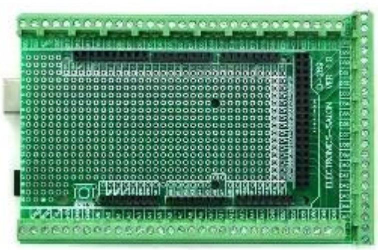
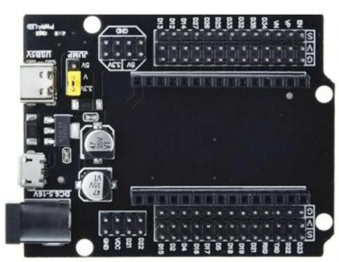
Fajarito, Yasme O.
BSCpE 3-1

Fruelda, John Aeron P.
BSCpE 3-1

Livelo, Ruzzel A.
BSCpE 3-1

APPENDIX C

Project Components

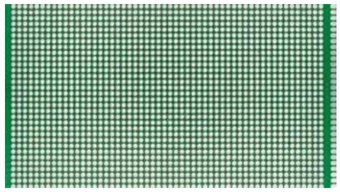
COMPONENTS	DESCRIPTION
	**Arduino Mega 2560** Serves as the main microcontroller of the system. It handles overall system control, sensor data processing, actuator control, and communication between different hardware components. Its large number of input/output pins makes it suitable for complex and multi-sensor applications.
	**Arduino Mega Terminal Board** Used as an interface board for the Arduino Mega 2560. It simplifies wiring connections by providing clearly labeled terminals, improving organization, reliability, and ease of maintenance of the system.
	**ESP32 with Expansion Board (Type-C)** Functions as the wireless communication module of the system. It enables Wi-Fi connectivity, allowing remote monitoring and control. The expansion board provides additional ports and stable power connections for peripherals.

7-Inch Nextion LCD Acts as the main human-machine interface (HMI) of the system. It displays real-time data such as temperature, smoke level, and system status, and allows users to monitor system parameters through its touchscreen interface.

Electric Stove (2 Units) Serves as the primary heating elements of the system. These generate the required heat for the smoking or heating process and are controlled automatically based on system settings.

Solid State Relay 40A (2 Units) Used to safely switch and control high-power electrical loads such as the electric stoves. It allows low-power control signals from the microcontroller to manage high-current AC loads efficiently and safely.

LM2596 Step-Down Converter (2 Units) Functions as a DC-DC buck converter that reduces higher input voltages to a stable lower output voltage. It is used to provide appropriate voltage levels for different electronic components.



9×15 Universal PCB (Green) Used for mounting and soldering electronic components. It provides a stable platform for circuit assembly and helps organize wiring connections within the system.

BTS7960 43A H-Bridge Motor Driver Controls the direction and speed of high-current DC motors or linear actuators. It enables forward and reverse motion and provides protection against overcurrent conditions.

12V Linear Actuator Used as a mechanical actuator to produce linear motion. It is controlled by the motor driver to automate physical movements such as opening, closing, or positioning mechanisms in the system.

2-Pin Terminal Blocks (30 Pieces) Used for secure and organized wire connections. These terminal blocks improve safety, simplify troubleshooting, and make system wiring more manageable.

MQ135 Smoke Sensor (2 Units) Detects smoke and air quality levels. It is used to monitor smoke concentration during operation and provides sensor data to the microcontroller for system safety monitoring.

DS3231 RTC Module Serves as a real-time clock that keeps accurate date and time information. It is used for time-based operations, scheduling, and data logging.

DHT22 Temperature Sensor Measures ambient temperature (and humidity, if required). It provides accurate temperature readings used for system monitoring and automatic control.

AWG #22 Stranded Wire – Black (1 Roll) Used for signal and low-current connections. The black wire is typically used for ground or negative connections.

AWG #16 Stranded Wire (15 Meters) Used for higher-current applications such as power lines for heating elements and motors. Its thicker gauge ensures safe and efficient current flow.

Jumper Wires Male-to-Female Used for temporary and flexible connections between modules, sensors, and microcontrollers during prototyping and testing.

Jumper Wires Male-to-Male Used for interconnecting components on breadboards or PCB headers during circuit development.

12V High-Pressure Water Pump Used to deliver water at high pressure for system functions that require fluid control, such as cooling, cleaning.

5V Relay Module Acts as an electromechanical switch that allows the microcontroller to control external devices operating at higher voltages or currents.

12V 10A Switching Power Supply Provides a stable 12V DC output to power system components such as actuators, pumps, and motor drivers. It ensures reliable and continuous operation.

220V Plug Used to connect the system to the main AC power source.

Shrinkable Tube Used for insulating and protecting electrical connections. It improves safety and prevents short circuits.

Heat-Resistant Wire (3 Meters) Used for high-temperature areas, particularly near heating elements. under elevated temperatures. It ensures safe operation.

APPENDIX D

Notice of Title Acceptance

CERTIFICATION

The undersigned members comprising the panel for oral nation hereby approve the Design Project entitled **SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production** including its team members composed of Denmark G. Acosta, Jayperson M. Caya, Yasme O. Fajarito, Ruzzel A. Livelo, and John Aeron P. Fruelda.

Evelyn A. Rodriguez, *DIT*
Design Project Adviser

ALONA L. LABAGUIS, *MSCpE*
Program Chairperson

JOHN EDGAR S. ANTHONY, *DIT*
Dean, College of Computer Studies

APPENDIX E

Endorsement for Final Defense

DESIGN PROJECT FINAL DEFENSE ENDORSEMENT

This Design Project entitled **"SMOKEFIN: AN IOT-ENABLED FISH SMOKING SYSTEM FOR SUSTAINABLE TINAPA PRODUCTION"** prepared and submitted by: **DENMARK G. ACOSTA, JAYPERSON M. CAYA, YASME O. FAJARITO, RUZZEL A. LIVELO, AND JOHN AERON P. FRUELDA** as a partial fulfillment of the requirement for the degree Bachelor of Science in Computer Engineering is hereby accepted and recommended for FINAL ORAL DEFENSE by me as their Design Project Adviser after a thorough review of their manuscript and system output. Their oral presentation may be scheduled on the MAY 06,2026.

Evelyn A. Rodriguez, *DIT*
Design Project Adviser

ALONA L. LABAGUIS, *MSCpE*
Program Chairperson

JOHN EDGAR S. ANTHONY, *DIT*
Dean, College of Computer Studies

APPENDIX F

Certificate of Originality

CERTIFICATION OF ORIGINALITY

This is to certify that the research work presented in this Design Project, **SMOKEFIN: AN IOT-ENABLED FISH SMOKING SYSTEM FOR SUSTAINBLE TINAPA PRODUCTION** for the degree **Bachelor of Science in Computer Engineering at the Mindoro State University** embodies the result of original and scholarly work contains words and ideas taken from published sources or written works that had been accepted as a basis for the award of a degree from any other higher education institution, except where proper referencing and acknowledgment were made.

DENMARK G. ACOSTA
Researcher

JAYPERSON M. CAYA
Researcher

YASME O. FAJARITO
Researcher

JOHN AERON P. FRUELDA
Researcher

RUZZEL A. LIVELO
Researcher

APPENDIX G

Grammarian's Certificate

GRAMMARIAN'S CERTIFICATE

This is to certify that the undersigned has reviewed the grammatical construction and the text organization of this Design Project entitled "SMOKEFIN: AN IOT-ENABLED FISH SMOKING SYSTEM FOR SUSTAINBLE TINAPA PRODUCTION".

Signed:

LYRIES K'ZANDRA G. ANGELES
Grammarian

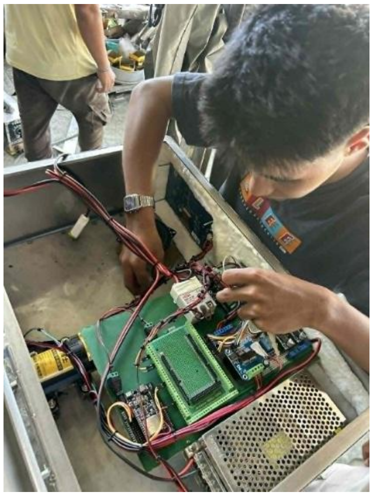
Confirmed:

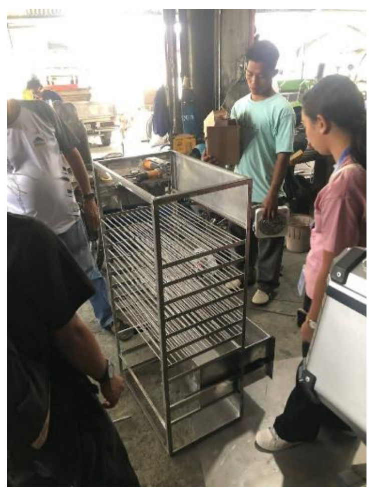
YASME O. FAJARITO
Project Technical Writer

JOHN AERON P. FRUELDA
Project Technical Writer

APPENDIX H

Design Project During Development





APPENDIX I

Design Project Evaluation Form

Design Project Title: SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production

Developers: Acosta, Denmark G.
Caya, Jayperson M.
Fajarito, Yasme O.
Fruelda, John Aeron P.
Livelo, Ruzzel A.

NAME:

____

Directions: Evaluate the design project using the rating scale
below. Check the box that corresponds to your rating.

Ratings:

4 – Acceptable 3 – Slightly Acceptable
2 – Not Acceptable 1 – Very Not Acceptable

1. Hygienic Design of the Physical Machine				
1. The machine is made of safe, food-grade materials that will not contaminate the fish.				
2. The inside of the chamber is easy to clean after each use.				
3. The smoke box keeps ash and dirt away from the fish.				
4. The design of the machine prevents pests or dust from getting inside.				

5. The fish racks keep and hold the product hygienic.				
AVERAGE				

2. System Functionality				
5. The system successfully performs the fish smoking process.				
6. The sensors accurately read the heat and smoke levels				
7. The linear actuator closes the smoke box properly when needed.				
8. The system operates consistently during the smoking process.				
9. The system maintains the desired temperature and smoke level automatically.				
AVERAGE				

3. Usability				
3.1. The LCD screen of the machine is easy to understand and use.				
3.2. The e-data such as temperature, smoke level, and time on the LCD screen are easy to read.				
3.3. The wood dust is easy to put into the machine manually.				
3.4. Fish racks are easy to arrange inside the machine.				
3.5. The overall operation of the machine is easy to learn and use.				
AVERAGE				

`

4. EFFICIENCY				
4.1. The smoke and heat spread evenly across all fish racks.				
4.2. The smoking duration is efficient and reasonable.				
4.3. The system improves productivity compared to traditional methods.				
4.4. The quality of the smoked fish (Tinapa) is very satisfactory.				
4.5. The machine minimizes fuel or material consumption during operation.				
AVERAGE				

5. Operational Safety				
5.1. Electrical and heating components are properly enclosed.				
5.2. Safety features such as breakers, and emergency stop buttons are easy to reach.				
5.3. All the wiring is safe and protected from water or heat.				
5.4. Sound alerts are loud enough to hear when there is a problem.				
5.5. The heat inside the machine does not burn the outer casing. The outer casing is safe to touch.				
AVERAGE				

6. Mobile Application Functionality				
[8] The mobile app accurately displays real-time data such as temperature, smoke level, and smoking duration from the machine.				

[9] The app responds quickly and reliably when refreshing or updating monitoring data				
[10] The app consistently connects to the system without errors or disconnections.				
[11] All features and buttons are working according to their label.				
[12] The app sends a clear notification when the fish is done				
AVERAGE				

7. Usability				
8.1. The mobile app is easy to navigate, even for first-time users.				
8.2. The interface design is clear, organized, and user-friendly.				
8.3. The information displayed on the application is easy to understand.				
8.4. The app responds quickly to user inputs such as opening screens, and refreshing data.				
8.5. The app layout helps users easily find important features and data.				
AVERAGE				

Remarks or suggestions:

____

____

____

Evaluator's Signature: _________________

Date: ______________________

APPENDIX J

Design Project Waiver

We, **DENMARK G. ACOSTA, JAYPERSON M. CAYA, YASME O. FAJARITO, JOHN AERON P. FRUELDA, RUZZEL A. LIVELO** of legal ages, the sole researchers of the Design Project entitled **"SMOKEFIN: AN IOT-ENABLED FISH SMOKING SYSTEM FOR SUSTAINABLE TINAPA PRODUCTION"**, hereby authorize our Adviser, Course Facilitator, and the College of Computer Studies - Research Unit to submit our Design Project for paper presentation, publication, and funding.

Whereas, if the research project will be accepted in any form of publication, or presentation, the researchers shall be assisted by their Adviser, Course Facilitator, and/or CCS Research Unit Head may represent the group.

Whereas the Adviser, Course Facilitator, and/or CCS Research Unit Head may improve the content of the manuscript by the requirements set forth by the conference and forum organizers. If improvements had been made by the Adviser, Course Facilitator, and/or CCS Research Unit Head, they would be a part of the research group as Co-Authors. The original researchers shall be notified and included as the researchers of the Research/Design Project.

Whereas our actions and decisions will be beneficial/advantageous for all of us, at the College of Computer Studies and its respective programs.

DENMARK G. ACOSTA	JAYPERSON M. CAYA
Project Developer	Project Developer
Date: ____________	Date: ____________

YASME O. FAJARITO
Project Developer
Date: ____________

JOHN AERON P. FRUELDA
Project Developer
Date: ____________

RUZZEL A. LIVELO
Project Developer
Date: ____________

APPENDIX K

Project Evaluation Photos

APPENDIX L

Certification of Oral Presentation

APPENDIX M

Matrix for Summary of Recommendations
Final Defense Compliance Form

Title oof the Study: **SMOKEFIN: AN IOT-ENABLED FISH SMOKING SYSTEM FOR SUSTAINABLE TINAPA PRODUCTION**

Course/Program: **BACHELOR OF SCIENCE IN COMPUTER ENGINEERING**

Date of Final Defense: **May 11, 2026**

Date of Submission of Revised Manuscript:

____

We, the undersigned members of the panel, hereby acknowledge that we have reviewed the final revised manuscript of the above-mentioned study. We confirm that the proponents have duly incorporated all suggestions and recommendations provided during the final defense, and the manuscript is now compliant with the required academic standards.

Panel Confirmation and Signatures

Name of Panelist	Signature	Date	Remarks
ENGR. ALONA L. LABAGUIS Chairman	___________	_________	__________
DR. EVELYN A RODRIGUEZ Member	___________	_________	__________
ENGR. ARNOLD F. REANO Member	___________	_________	__________

Prepared by:

____

Signature: ____________________ Date: ______________

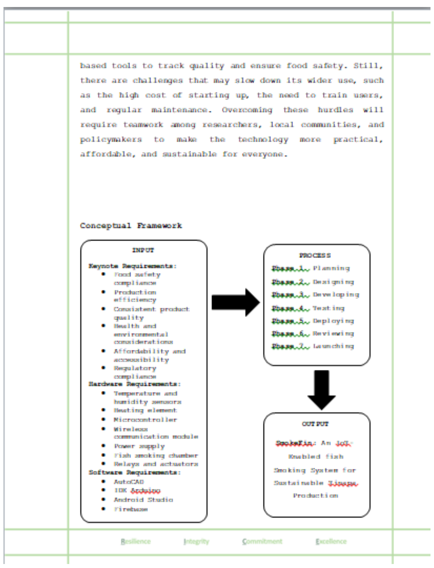
List of Recommendation and Suggestions

No.	RECOMMENDATION/SUGGESTION	ACTION TAKEN	PAGE NO.
1	Revise the Conceptual Framework diagram to follow the correct format where all components are aligned horizontally.	Adjusted the Conceptual Framework diagram to ensure all elements are properly aligned in one row.	20
2	Improve the Block Diagram for better clarity and understanding.	Revised the Block Diagram to make the process clearer and easier to understand.	27
3	Clarify the methodology description to indicate that it is intended for the application.	Edited the methodology description to clearly specify that it is for the application.	33
4	Revise the methodology description to indicate that it is intended for the hardware/machine.	Modified the methodology description to clearly specify that it is for the hardware or machine.	34
5	Correct and organize the table to follow proper format.	Fixed and reformatted the table to ensure it is accurate and properly arranged.	62
6	Improve the conclusions section.	Revised and refined the conclusions to	71

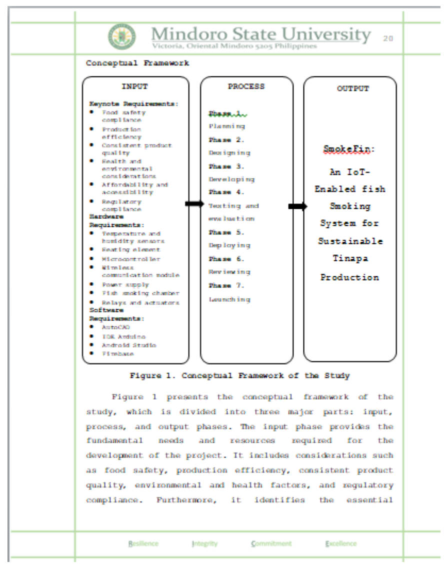
		make them clearer and more appropriate.	

No 1 Recommendation/ Suggestion

Adjusted the Conceptual Framework diagram to ensure all elements are properly aligned in one row.


based tools to track quality and ensure food safety. Still, there are challenges that may slow down its wider use, such as the high cost of starting up, the need to train users, and regular maintenance. Overcoming these hurdles will require teamwork among researchers, local communities, and policymakers to make the technology more practical, affordable, and sustainable for everyone.

Conceptual Framework



Conceptual Framework

Figure 1. Conceptual Framework of the Study

Figure 1 presents the conceptual framework of the study, which is divided into three major parts: input, process, and output phases. The input phase provides the fundamental needs and resources required for the development of the project. It includes considerations such as food safety, production efficiency, consistent product quality, environmental and health factors, and regulatory compliance. Furthermore, it identifies the essential

No 2 Recommendation/ Suggestion

Revised the Block Diagram to make the process clearer and easier to understand.

Block Diagram

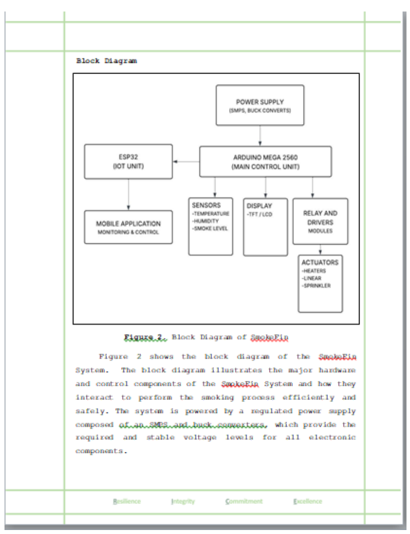


Figure 2. Block Diagram of SmokeFin

Figure 2 shows the block diagram of the SmokeFin System. The block diagram illustrates the major hardware and control components of the SmokeFin System and how they interact to perform the smoking process efficiently and safely. The system is powered by a regulated power supply composed of an SMPS and buck converters, which provide the required and stable voltage levels for all electronic components.

Block Diagram

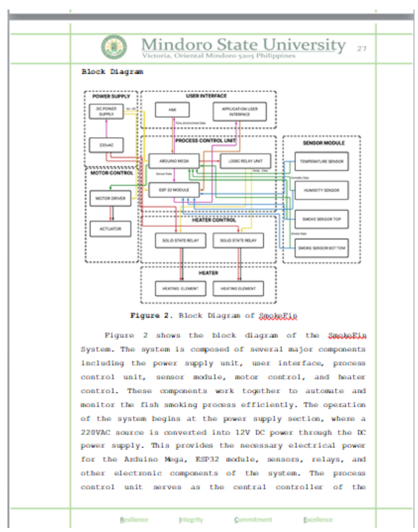


Figure 2. Block Diagram of SmokeFin

Figure 2 shows the block diagram of the SmokeFin System. The system is composed of several major components including the power supply unit, user interface, process control unit, sensor module, motor control, and heater control. These components work together to automate and monitor the fish smoking process efficiently. The operation of the system begins at the power supply section, where a 220VAC source is converted into 12V DC power through the DC power supply. This provides the necessary electrical power for the Arduino Mega, ESP32 module, sensors, relays, and other electronic components of the system. The process control unit serves as the central controller of the

No 3 Recommendation/ Suggestion

Edited the methodology description to clearly specify that it is for the application.


control. Each sprint included testing and validation, followed by refinements based on feedback from fish processors, technical experts, and advisers to ensure that the system addressed both technical objectives and user needs. This iterative process allowed the researchers to quickly respond to challenges such as connectivity limitations, sensor accuracy, and food safety considerations, ensuring continuous improvement of the prototype. By applying Agile, the project achieved a reliable, user-friendly, and sustainable system that modernizes traditional Tinapa production through automation, real-time monitoring, and improved food quality.

Figure 6: Rapid Application Development

For the development of the SmokeFin mobile application, the developers decided to use Rapid Application Development Methodology. The process starts with Analysis and Quick Design, where the developers identified the core requirements for real-time monitoring of temperature, humidity, and smoke levels. During this phase, flowcharts and conceptual blueprints were rapidly created to map out how the app would interface with the ESP32 microcontroller and the

Resilience Integrity Commitment Excellence


Mindoro State University 34
Victoria, Oriental Mindoro 5205 Philippines

application integration, and heating and ventilation control. Each sprint included testing and validation, followed by refinements based on feedback from fish processors, technical experts, and advisers to ensure that the system addressed both technical objectives and user needs. This iterative process allowed the researchers to quickly respond to challenges such as connectivity limitations, sensor accuracy, and food safety considerations, ensuring continuous improvement of the prototype. By applying Agile, the project achieved a reliable, user-friendly, and sustainable system that modernizes traditional Tinapa production through automation, real-time monitoring, and improved food quality.

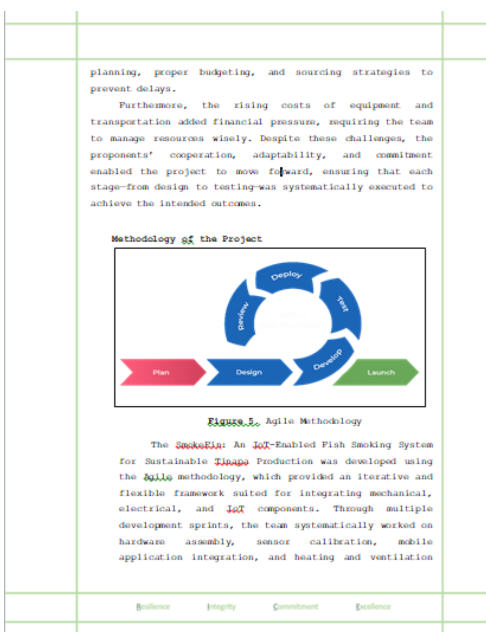
Figure 6: Rapid Application Development

For the development of the SmokeFin mobile application, the developers decided to use Rapid Application Development Methodology. The process starts with Analysis and Quick Design, where the developers identified the core requirements for real-time monitoring of temperature, humidity, and smoke levels. During this phase, flowcharts and conceptual blueprints were rapidly created to map out how the app would interface with the ESP32 microcontroller and the

Resilience Integrity Commitment Excellence

No 4 Recommendation/ Suggestion

Modified the methodology description to clearly specify that it is for the hardware or machine.


planning, proper budgeting, and sourcing strategies to prevent delays.

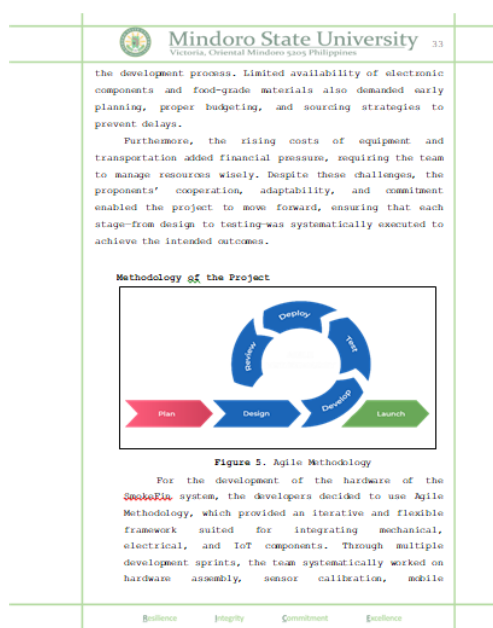
Furthermore, the rising costs of equipment and transportation added financial pressure, requiring the team to manage resources wisely. Despite these challenges, the proponents' cooperation, adaptability, and commitment enabled the project to move forward, ensuring that each stage—from design to testing—was systematically executed to achieve the intended outcomes.

Methodology of the Project

Figure 5. Agile Methodology

The SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production was developed using the Agile methodology, which provided an iterative and flexible framework suited for integrating mechanical, electrical, and IoT components. Through multiple development sprints, the team systematically worked on hardware assembly, sensor calibration, mobile application integration, and heating and ventilation

Resilience Integrity Commitment Excellence


Mindoro State University 33
Victoria, Oriental Mindoro 5205 Philippines

the development process. Limited availability of electronic components and food-grade materials also demanded early planning, proper budgeting, and sourcing strategies to prevent delays.

Furthermore, the rising costs of equipment and transportation added financial pressure, requiring the team to manage resources wisely. Despite these challenges, the proponents' cooperation, adaptability, and commitment enabled the project to move forward, ensuring that each stage—from design to testing—was systematically executed to achieve the intended outcomes.

Methodology of the Project

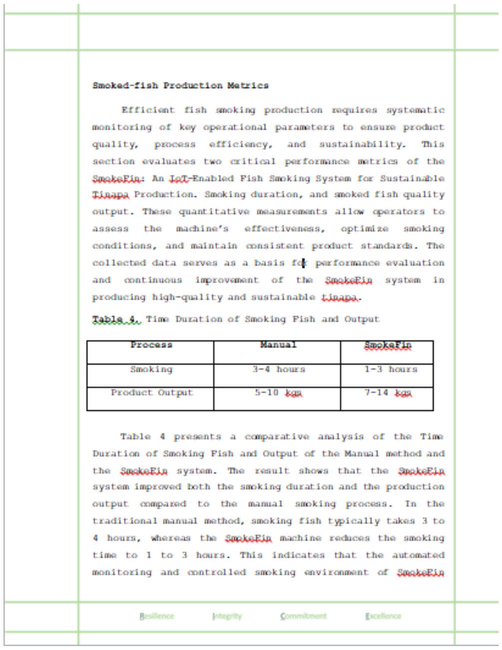
Figure 5. Agile Methodology

For the development of the hardware of the SmokeFin system, the developers decided to use Agile Methodology, which provided an iterative and flexible framework suited for integrating mechanical, electrical, and IoT components. Through multiple development sprints, the team systematically worked on hardware assembly, sensor calibration, mobile

Resilience Integrity Commitment Excellence

No 5 Recommendation/ Suggestion

Fixed and reformatted the table to ensure it is accurate and properly arranged.



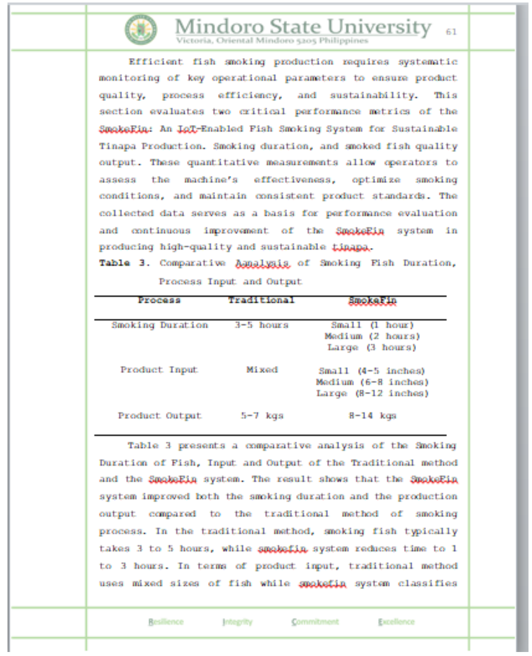
Smoked-fish Production Metrics

Efficient fish smoking production requires systematic monitoring of key operational parameters to ensure product quality, process efficiency, and sustainability. This section evaluates two critical performance metrics of the SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production. Smoking duration, and smoked fish quality output. These quantitative measurements allow operators to assess the machine's effectiveness, optimize smoking conditions, and maintain consistent product standards. The collected data serves as a basis for performance evaluation and continuous improvement of the SmokeFin system in producing high-quality and sustainable tinapa.

Table 4. Time Duration of Smoking Fish and Output

Process	Manual	SmokeFin
Smoking	3-4 hours	1-3 hours
Product Output	5-10 kgs	7-14 kgs

Table 4 presents a comparative analysis of the Time Duration of Smoking Fish and Output of the Manual method and the SmokeFin system. The result shows that the SmokeFin system improved both the smoking duration and the production output compared to the manual smoking process. In the traditional manual method, smoking fish typically takes 3 to 4 hours, whereas the SmokeFin machine reduces the smoking time to 1 to 3 hours. This indicates that the automated monitoring and controlled smoking environment of SmokeFin



Efficient fish smoking production requires systematic monitoring of key operational parameters to ensure product quality, process efficiency, and sustainability. This section evaluates two critical performance metrics of the SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production. Smoking duration, and smoked fish quality output. These quantitative measurements allow operators to assess the machine's effectiveness, optimize smoking conditions, and maintain consistent product standards. The collected data serves as a basis for performance evaluation and continuous improvement of the SmokeFin system in producing high-quality and sustainable tinapa.

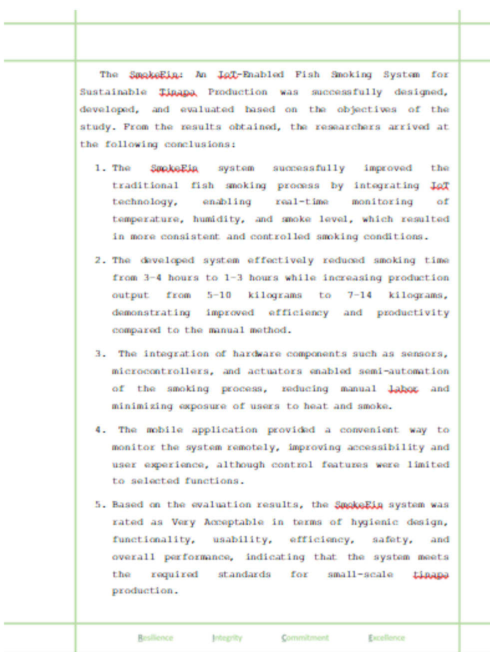
Table 3. Comparative Aanalysis of Smoking Fish Duration, Process Input and Output

Process	Traditional	SmokeFin
Smoking Duration	3-5 hours	Small (1 hour) Medium (2 hours) Large (3 hours)
Product Input	Mixed	Small (4-5 inches) Medium (6-8 inches) Large (8-12 inches)
Product Output	5-7 kgs	8-14 kgs

Table 3 presents a comparative analysis of the Smoking Duration of Fish, Input and Output of the Traditional method and the SmokeFin system. The result shows that the SmokeFin system improved both the smoking duration and the production output compared to the traditional method of smoking process. In the traditional method, smoking fish typically takes 3 to 5 hours, while smokefin system reduces time to 1 to 3 hours. In terms of product input, traditional method uses mixed sizes of fish while smokefin system classifies

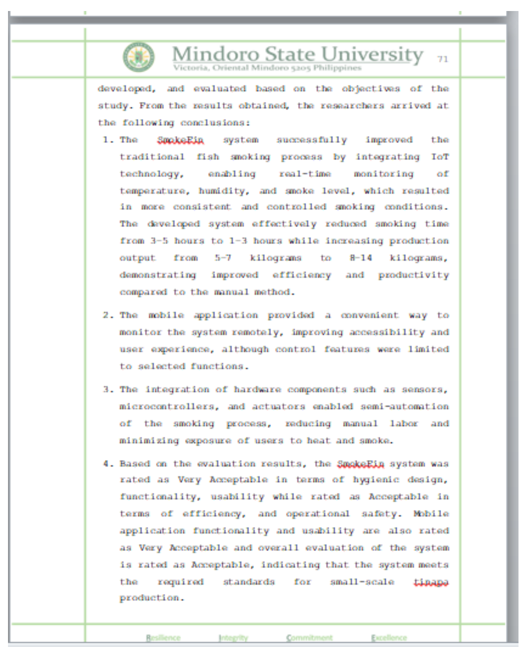
No 6 Recommendation/ Suggestion

Revised and refined the conclusions to make them clearer and more appropriate.



The SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production was successfully designed, developed, and evaluated based on the objectives of the study. From the results obtained, the researchers arrived at the following conclusions:

1. The SmokeFin system successfully improved the traditional fish smoking process by integrating IoT technology, enabling real-time monitoring of temperature, humidity, and smoke level, which resulted in more consistent and controlled smoking conditions.
2. The developed system effectively reduced smoking time from 3-4 hours to 1-3 hours while increasing production output from 5-10 kilograms to 7-14 kilograms, demonstrating improved efficiency and productivity compared to the manual method.
3. The integration of hardware components such as sensors, microcontrollers, and actuators enabled semi-automation of the smoking process, reducing manual labor and minimizing exposure of users to heat and smoke.
4. The mobile application provided a convenient way to monitor the system remotely, improving accessibility and user experience, although control features were limited to selected functions.
5. Based on the evaluation results, the SmokeFin system was rated as Very Acceptable in terms of hygienic design, functionality, usability, efficiency, safety, and overall performance, indicating that the system meets the required standards for small-scale tinapa production.



developed, and evaluated based on the objectives of the study. From the results obtained, the researchers arrived at the following conclusions:

1. The SmokeFin system successfully improved the traditional fish smoking process by integrating IoT technology, enabling real-time monitoring of temperature, humidity, and smoke level, which resulted in more consistent and controlled smoking conditions. The developed system effectively reduced smoking time from 3-5 hours to 1-3 hours while increasing production output from 5-7 kilograms to 8-14 kilograms, demonstrating improved efficiency and productivity compared to the manual method.
2. The mobile application provided a convenient way to monitor the system remotely, improving accessibility and user experience, although control features were limited to selected functions.
3. The integration of hardware components such as sensors, microcontrollers, and actuators enabled semi-automation of the smoking process, reducing manual labor and minimizing exposure of users to heat and smoke.
4. Based on the evaluation results, the SmokeFin system was rated as Very Acceptable in terms of hygienic design, functionality, usability while rated as Acceptable in terms of efficiency, and operational safety. Mobile application functionality and usability are also rated as Very Acceptable and overall evaluation of the system is rated as Acceptable, indicating that the system meets the required standards for small-scale tinapa production.

APPENDIX N

IEEE Format

Denmark G. Acosta
Mindoro State University
College of Computer Studies
Oriental Mindoro, Philippines
denmarkgalacanacosta@gmail.com

Jayperson M. Caya
Mindoro State University
College of Computer Studies
Oriental Mindoro, Philippines
jaypersonmarasigancaya@gmail.com

Yasme O. Fajarito
Mindoro State University
College of Computer Studies
Oriental Mindoro, Philippines
fajaritoyasme@gmail.com

John Aeron P. Fruelda
Mindoro State University
College of Computer Studies
Oriental Mindoro, Philippines
johnaeronfruelda@gmail.com

Ruzzel A. Livelo
Mindoro State University
College of Computer Studies
Oriental Mindoro, Philippines
ruzzellivelo@gmail.com

Dr. Evelyn A Rodriguez
Mindoro State University
College of Computer Studies
Oriental Mindoro, Philippines
allabaguis.0904@gmail.com

Abstract. The SmokeFin project is a technological modernization of the traditional Filipino fish smoking process (Tinapa), developed to address the limitations of labor-intensive, inconsistent, and health-risking manual methods. This IoT-enabled fish smoking system integrates sensors for temperature, humidity, and smoke level monitoring, combined with a microcontroller-based architecture utilizing an Arduino Mega 2560 and ESP32 module for real-time data processing and wireless communication. The system enhances the smoking process through controlled heating elements, actuators, and a user-friendly mobile application for remote monitoring. Performance evaluation was conducted in accordance with ISO 14159:2002 for hygienic machine design and ISO 25010:2023 for software quality, assessing criteria such as functionality, usability, efficiency, safety, and reliability. Results showed that SmokeFin reduces smoking duration from 3-5 hours to 1-3 hours while increasing production output from 5-7 kg to 8-14 kg per batch. The system received an overall evaluation rating of "Very Acceptable," highlighting its effectiveness in improving operational safety, consistency, and user experience. Ultimately, SmokeFin provides a safer, more efficient, and sustainable solution for small-scale fish processors and vendors, supporting food security, economic growth, and technological advancement while preserving the cultural practice of traditional Tinapa production.

Keywords: IoT, Fish Smoking System, Tinapa, SmokeFin, Automation, Real-Time Monitoring, Food Processing Technology, Sustainable Production.

1. Introduction

Fish plays an important role in the lives of many people living in coastal and riverside communities. It is not only a daily source of food but also a major source of livelihood for small-scale fishers and their families. However, fresh fish spoils quickly, especially in areas without access to refrigeration to address this, Filipinos have traditionally turned to smoking fish, producing what is commonly known as Tinapa. This method helps extend the shelf life of fish while adding flavor and value, allowing communities to store, sell, and transport their catch more effectively.

Traditionally, fish smoking is done manually over open fires. This process takes a lot of time, requires constant monitoring, and exposes the fish processors often women and elders to smoke, heat, and other health hazards. While this method has been passed down for generations, it can be inconsistent and labor intensive. To improve this practice, a SmokeFin machine was developed. This machine is designed to make the smoking process safer, faster, and more consistent, helping small fish vendors produce higher quality products while reducing health risks and physical strain [1].

The development of an Internet of Things (IoT) based Tinapa machine aims to enhance fish smoking by automating the process and providing real-time data on temperature, humidity, and smoking duration. This system combines sensors with wireless technology to help improve quality, reduce waste, and ensure safer food production [2]. The researchers chose Adame Jr.'s Tinapahan located at Barangay San Isidro, Bongabong Oriental Mindoro as beneficiary which is one of the suppliers of Tinapa product in the town. Adame Jr.'s Tinapahan currently using traditional way of making Tinapa which requires manual process of smoking. Researchers believed that the project would bring benefit to their business and help them keep up with the advancement of technology. This innovation is also helpful for community cooperatives and small businesses who rely on Tinapa for income.

In addition, this project supports sustainable development goals such as SDG 2 known as Zero hunger, as the machine helps reduce food waste and extend fish shelf life, ensuring a more stable and reliable food supply for fish-based communities. It also supports SDG 8 known as Decent Work and Economic Growth, by improving efficiency, and safety in tinapa production, the system increases productivity and opportunities for local tinapa vendors. Moreover, the project also supports SDG 9 known as Industry, Innovation, and Infrastructure by integrating IoT technology into traditional fish process, promoting innovation, modernizing local industries, and strengthening sustainable production infrastructure. The IoT-enabled Tinapa maker not only preserves fish but also upholds culture, supports local livelihoods, and builds a more sustainable future for fish-based communities.

Lastly, this project aligns with ISO standards to ensure safety, quality, and sustainability. ISO 14159:2002 known as safety of the Machinery – Hygiene Requirements for the design was integrated to ensure hygienic conditions were the machine consistently maintained during operation. Furthermore, this standard guided the design of pre-processing, processing, and post-processing stages, emphasizing cleanability and contamination prevention. By adhering to ISO principles, the design minimized hygiene risks by enabling effective removal of residues and microorganisms, thereby ensuring product safety and quality [3].

To further support the software quality assurance of the project's mobile application, ISO 2501025010:2011 known as Systems and Software Quality Requirements and Evaluation standard was integrated. This provides a comprehensive framework for identifying software strengths and weaknesses, allowing developers to systematically assess application performance based on both functional and non-functional requirements. By integrating this standard into the evaluation process, the developers were able to ensure that the application met user expectations while maintaining high standards of software reliability and operational efficiency [4].

The objectives of the study are as follows:

1. To design an IoT-based fish smoking system that can monitor environmental conditions such as temperature, humidity, and smoke level to achieve uniform smoking results;
2. To develop an interface that enables users to remotely monitor the fish smoking process through a mobile application.
3. To integrate electronic components such as sensors, heating elements, actuators, and wireless communication modules to support automated monitoring and controlled operation of the fish smoking process, while providing real-time system feedback through the machine's interface and mobile application; and
4. To test and evaluate the system's overall performance in accordance with the ISO 14159:2002 and ISO 25010:2023 standards.

1. Methodology

2.0 Project Analysis

Fish smoking has long been an important preservation method in the Philippines, particularly in coastal and riverside communities where fish serves as both a primary food source and a livelihood. However, traditional fish smoking methods are labor-intensive, health-risking, and often yield inconsistent product quality due to uneven temperature and smoke control. As demand for safer and more efficient food production continues to grow, there is a pressing need to modernize Tinapa (smoked fish) processing with innovative technologies.

The SmokeFin project was chosen to address the longstanding challenges of traditional fish smoking methods, which are labor-intensive, inconsistent, and hazardous to the health of small-scale processors. The primary goal of this project is to design and implement an IoT-enabled system that automates and monitors the Tinapa smoking process, ensuring improved efficiency, safety, and product quality. By integrating sensors, a microcontroller, and wireless communication, the system provides real-time data on temperature, humidity, and duration, allowing users to remotely monitor the smoking process. This methodology is significant because it combines food safety compliance, technological innovation, and sustainability to meet the pressing needs of fish-based communities. The chapter outlines the project objectives, the research foundation that guided the design, conceptualization of the solution, and the architecture of the SmokeFin system.

The problem definition highlights the need to modernize fish smoking practices, particularly in communities where Tinapa is a major livelihood. Traditional methods expose processors to harmful smoke and produce inconsistent results, leading to inefficiency and health risks. The SmokeFin project seeks to overcome these problems by creating an automated system that is affordable, user-friendly, and suitable for small-scale production. The design must meet key requirements such as accurate temperature and humidity monitoring, safe handling of smoke, food safety compliance, and energy efficiency. Constraints include cost limitations, environmental considerations, and the need for reliable connectivity, which were carefully reviewed and adjusted during the course of the project to refine the final set of system requirements.

A comprehensive background study was conducted to examine existing technologies and innovations in fish smoking. Traditional ovens such as the Chorkor and cylindrical ovens were analyzed alongside modern systems like electric kilns and IoT-enabled prototypes. Literature reviews also revealed gaps in consistency, sustainability, and scalability, which the SmokeFin project aims to address. Insights from existing research—such as the benefits of microcontroller-based automation and the potential of IoT for real-time monitoring—played a crucial role in shaping the system design. This grounding ensured that SmokeFin not only improves on traditional methods but also aligns with international food safety and environmental standards

The design conceptualization process involved brainstorming and evaluating different approaches to integrating sensors, actuators, and microcontrollers. The final design was chosen for its practicality, innovation, and close alignment with project goals. By automating heat control, smoke circulation, and monitoring, while also enabling user-friendly remote access, the selected concept balanced technical sophistication with affordability and ease of use for small-scale vendors.

The system architecture integrates several key components into a cohesive framework. Sensors such as K-Type Thermocouple sensor to collect environmental and process data from the smoking chamber. These inputs are processed by the ESP32 microcontroller, which controls actuators like the heating element, fan, and servo damper to regulate conditions inside the chamber. Safety mechanisms, including overtemperature cutoff and fire detection, add reliability to the process. Wireless connectivity ensures that data is transmitted and accessed through mobile applications, giving users full visibility and control. The system is powered by AC mains with battery backup and energy management components, ensuring stable and continuous operation. Altogether, the architecture demonstrates how IoT technology can transform traditional Tinapa production into a safer, and more sustainable process.

2.1Project Management

The development of the SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production required careful planning, coordination, and resource management to ensure its successful completion. Since the project involved the integration of mechanical components, electronic modules, and IoT technology, the proponents faced the challenge of balancing technical complexities with practical usability.

Effective communication and collaboration among team members were essential in addressing design issues, coordinating tasks, and making timely decisions throughout the development process. Limited availability of electronic components and food-grade materials also demanded early planning, proper budgeting, and sourcing strategies to prevent delays. Furthermore, the rising costs of equipment and transportation added financial pressure, requiring the team to manage resources wisely. Despite these challenges, the proponents' cooperation, adaptability, and commitment enabled the project to move forward, ensuring that each stage—from design to testing—was systematically executed to achieve the intended outcomes.

2.2 Project Requirement

2.2.1 Design and Development of SMOKEFIN

Materials	Description
Arduino Mega 2560	Serves as the main microcontroller of the system. It handles overall system control, sensor data processing, actuator control, and communication between different hardware components.
Arduo Mega Terminal Board	Used as an interface board for the Arduino Mega 2560. It simplifies wiring connections by providing clearly labeled terminals, reliability, and ease of maintenance of the system.
ESP32 with Expansion Board	Functions as the wireless communication module of the system. It enables Wi-Fi connectivity, allowing remote monitoring and control.
Nextion LCD	Acts as the main human-machine interface (HMI) of the system. It displays real-time data such as temperature, smoke level, and system status, and allows users to monitor system parameters through its touchscreen interface.
Electric Stove	Serves as the primary heating elements of the system. These generate the required heat for the smoking or heating process and are controlled automatically based on system settings.

Table 1. Project Components

The hardware components of SMOKEFIN include an **Arduino Mega 2560, Arduino Mega Terminal Board, ESP32 with Expansion Board (Type-C),7-Inch Nextion LCD, Electric Stove (2 Units),Solid State Relay 40A (2 Units, LM2596 Step-Down Converter (2 Units, 9×15 Universal PCB (Green),BTS7960 43A H-Bridge Motor Driver, 12V Linear Actuator, 2-Pin Terminal Blocks**l**, MQ135 Smoke sensor, DS3231 RTC Module,DHT22 Temperature Sensor,AWG #22 Stranded Wire – Black (1 Roll,AWG #16 Stranded Wire (15 Meters, Jumper Wires, 12V High-Pressure Water Pump, 5V Relay Module, 12V 10A Switching Power Supply, 220V Plug,Shrinkable Tube** and **Heat-**

Resistant Wire (3 Meters). On the software side, the system utilized Arduino IDE with Arduino Studio, AutoCAD, and Firebase.

2.3 Project Data Gathering

The researchers gathered information for their design project using various methods. They conducted in-person interviews, observed existing processes, and explored multiple websites to get inspiration and lean from other's experience.

2.4 Project Evaluation

A four-point Likert Scale was used to quantify system performance, with ratings ranging from 1 (Very Not Acceptable) to 4 (Very acceptable). The verbal interpretations were based on the corresponding statistical limits to ensure consistent evaluation across all criteria. Ten (10) respondents evaluated the design project using the mentioned rating scale, the metrics that was evaluated are Hygienic **Design of the Physical Machine, System Functionality, Usability, Efficiency, Operational Safety, Mobile Application Functionality, and Usability.**

2.5 Flowchart of SMOKEFIN

The system flowchart illustrates the overall operation of the SMOKEFIN monitoring system. It begins when the system is powered on using an AC power supply, initiating system initialization where the LCD screen, heating element, and temperature and humidity sensors are activated. After initialization, the LCD displays fish size options for the user, and once a selection is made, the fish is placed inside the smoking chamber to start the automatic smoking process. During operation, the temperature and humidity sensors continuously monitor conditions inside the chamber and display real-time data on the LCD. The user also has the option to manually stop the process at any time depending on their preferred level of doneness. Once the set smoking time is completed, the system displays the message "Smoke Fish | Process Done," indicating successful completion of the process.

3. Results and Discussion

3.1 Design and Development of SMOKEFIN

Figure 1. The **Front View of the Smokefin Machine**

Figure 1 shows the front right view of the SmokeFin machine, where the main external parts of the system can be clearly seen. The machine was first designed using 3D modeling software to visualize its structure, dimensions, and component placement before fabrication. The main frame was built using durable metal to ensure stability, followed by the installation of outer panels and glass doors for protection and monitoring. Racks were placed inside the chamber to hold fish products during operation. The smoke box was installed at the lower part of the machine to produce smoke from wood dust or other materials. A cable conduit was added to organize and protect electrical wiring, and the Nextion LCD display was mounted on the upper front section for easy monitoring and control of system parameters.

Figure 3. The Smoke Box of SmokeFin

Figure 3 shows the smoke box and its components of the SmokeFin machine. The smoke box serves as the main chamber where smoke is generated using wood dust. It was built with a durable metal enclosure to withstand high temperatures during operation. Heating elements were installed at the bottom to produce the heat needed for smoke generation, with insulation added around them to reduce heat loss and improve safety. A sprinkler system was also included as a safety feature to control excessive smoke or prevent overheating, connected through proper piping for smooth water flow. An ash box was placed at the lower portion to collect burned residues and designed to be removable for easy cleaning. All components were carefully arranged to ensure safe and efficient operation.

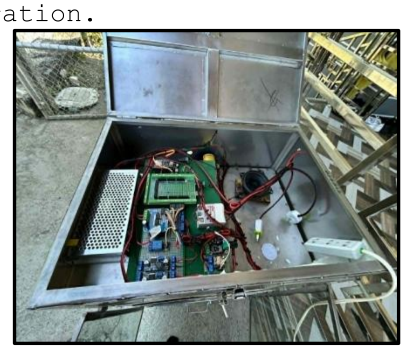


Figure 2. Wiring Box of the SmokeFin Machine

Figure 2 shows the internal wiringg connections of the SmokeFin machine. After developing the machine's casing and integrating its major features, the developers proceeded with the wiring and electrical connections of the system. Electrical components such as the sensors, power supply, relays, and microcontroller were carefully interconnected to ensure proper communication and functionality of the automated features. The wiring arrangement was organized systematically to promote safety, reliability, and efficient operation of the machine. Proper cable management and insulation were also applied to minimize the risk of short circuits and electrical hazards during operation. This stage of development was essential in ensuring that the SmokeFin system could perform its monitoring and automation processes accurately and consistently.

3.2 Mobile App Integration

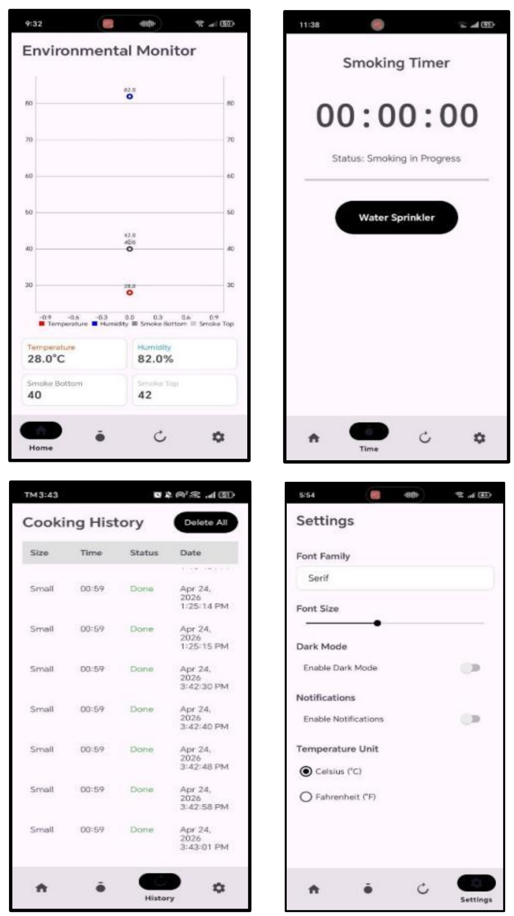


Figure 3. Mobile Application of SmokeFin

Figure 3 shows the mobile application monitoring system developed for the SmokeFin machine. The application was designed to provide remote monitoring and easier access to the machine's operational status through a smartphone device. The development process started by planning the interface and features needed in the application, including temperature monitoring, smoke status, notifications, and system updates. After planning the design, the application was connected to the machine's control system through wireless communication technology to allow real-time data transmission between the SmokeFin machine and the mobile device. The necessary programming and integration were performed to ensure that the application could properly receive and display system information from the machine. User interface elements such as buttons, status indicators, and monitoring panels were arranged clearly to make the application simple and easy to use. The system was tested multiple times to verify connectivity, responsiveness, and accuracy of the displayed data. Necessary improvements were made to ensure reliable and convenient remote monitoring of the SmokeFin machine.

3.3 Testing and Evaluation

Evaluation Category	**Overall Mean**	**Verbal Interpretation**
Hygienic Design of the Machine	3.74	Very Acceptable
System Functionality	3.55	Acceptable
Usability	3.68	Very Acceptable
Efficiency	3.13	Acceptable
Operational Safety	3.18	Acceptable
Mobile Application Functionality	3.6	Very Acceptable
Mobile Application Usability	3.56	Very Acceptable
Average	3.49	Acceptable

Table 2. Results of System Evaluation

The overall results of the system evaluation indicate that the SMOKEFIN machine achieved a weighted mean of 3.49, interpreted as "Acceptable." Among the evaluation categories, Hygienic Design of the Machine obtained the highest rating (3.74), followed by Usability (3.68) and Mobile Application Functionality (3.60), all interpreted as "Very Acceptable," reflecting strong performance in cleanliness, ease of use, and mobile integration. Meanwhile, System Functionality (3.55), Efficiency (3.13), and Operational Safety (3.18) received "Acceptable" ratings, indicating satisfactory performance in core operations and safety. Overall, the results demonstrate that the SMOKEFIN machine is reliable, user-friendly, and meets acceptable standards across all evaluated aspects.

4.0 Conclusion

The study concludes that SmokeFin: An IoT-Enabled Fish Smoking System for Sustainable Tinapa Production successfully achieved its objective of developing an IoT-based fish smoking system that improves the traditional tinapa production process through real-time monitoring and semi-automation. By integrating sensors, microcontrollers effectively monitored temperature the system effectively monitored temperature, humidity, and smoke levels while providing users with convenient remote access and monitoring capabilities. The developed system significantly reduced smoking time from 3-5 hours to 1-3 hours and increased production capacity from 5-7 kilograms to 8-14 kilograms, demonstrating improved efficiency, productivity, and consistency compared to the conventional manual method. Furthermore, the integration of automated hardware components minimized manual labor and reduced users' exposure to excessive heat and smoke during operation. Evaluation results revealed that the system obtained "Very Acceptable" ratings in terms of hygienic design, functionality, and usability, while receiving "Acceptable" ratings in efficiency and operational safety. Overall, the SmokeFin system was evaluated as Acceptable, confirming its reliability, practicality, and suitability for small-scale sustainable tinapa production despite minor limitations in automation and connectivity features.

5.0 Recommendations

Future improvements to the SmokeFin system may include the implementation of fully automated control mechanisms for temperature, humidity, and smoke levels to further enhance efficiency and consistency in the smoking process. Additional features such as fish readiness detection using sensors or image processing technology may also be integrated to reduce dependence on manual observation and improve product quality. Enhancing the mobile application with full remote-control capabilities, real-time notifications, and alert systems for abnormal conditions is also recommended to improve user convenience and operational safety. To ensure continuous functionality, offline operation and local network connectivity may be incorporated for areas with unstable internet access. Future researchers may also focus on scaling the system for commercial-level production while maintaining efficiency, hygiene, and consistency. Moreover, the integration of alternative energy sources such as solar power, together with improved structural materials, fire detection mechanisms, and automatic shutdown features, is recommended to enhance sustainability, durability, and overall system safety for long-term utilization.

BIBLIOGRAPHY

[1] R. M. Davies and O. A. Davies, "Traditional and Improved Fish Processing Technologies in Bayelsa State, Nigeria," European Journal of Scientific Research, vol. 26, no. 4, pp. 539-548, 2009.

[2] O. M. Osineye et al., "Comparative Evaluation of Performances of Biomass-Sourced Briquette (Sawdust), Wood and Charcoal as Fuelling Materials in Fish Smoking," Journal of Sustainable Development, vol. 17, no. 1 & 2, pp. 106-111, Sep. 2020.

[3] International Organization for Standardization. (2002). ISO 14159:2002 Safety of machinery—Hygiene requirements for the design of machinery. Available Online: https://www.iso.org/standard/23150.html

[4] Suryadi, D. A., & Sulistiyani, E. (2022). Evaluation of information quality using ISO/IEC 25010:2011 (Case research: Menu Harianku application). International

Journal of Innovation in Enterprise System, 6(2), 95-102.

[5] A. M. Djordjevic, "Smoke and Smoked Fish Production," Meat Technology, vol. 61, no. 2, pp. 93-102, 2020.

[6] and Sampurno, "Design of mobile, portable and ergonomic fish smoke machine on a motorcycle," IOP Conference Series: Materials Science and Engineering, vol. 674, no. 1, p. 012047, Nov. 2019

[7] M. I. Joesidawati, Suwarsih, A. W. Nuruddin, and Sriwulan, "Performance Test Smoked Tool Fish Which Effective, Hygienic, and Eco-Friendly," Revista de Gestão Social e Ambiental, vol. 18, no. 6, p. e07478, Jun. 2024, doi: 10.24857/rgsa. v18n6-119.

[8] E. Fradinata and Z. M. Kesuma, "Design of Fishing Smoke (Saltization) Equipment and Its Process for Improving the Economy Community in Lampulo Banda Aceh," J. Pengabdian Pada Masyarakat, vol. 5, no. 1, pp. 213-220, Feb. 2020. doi:10.30653/002.202051.250

[9] Rasco, B. Smoking Fish at Home Safely, PNW238. Pacific Northwest Extension publications— Washington State University Extension. 2009. Available online: http://www.uaf.edu

[10] Tsai, W.T.; Lee, M.K.; Chang, Y.M. Fast pyrolysis of rice straw, sugarcane bagasse and coconut shell in an induction-heating reactor. J. Anal. Appl. Pyrolysis 2006, 76, 230-237.

[11] A. J. R. Taduran, "The Development of a Microcontroller based Smoked Fish Machine," International Journal of Engineering Trends and Technology, vol. 72, no. 3, pp. 370-380, Mar. 2024, doi: 10.14445/22315381/IJETTV72I3P132 .

[12] Mendoza, Xavier & Abug, Nerisa. (2025). Tinapa Production: An Analysis of Current Practices, Challenges, and Potential for Enhancement in the Province of Cavite. 1-13. 10.17613/a2ryj-wrx87.

[13] Pramanik, Sabyasachi, Soma Bag, Atanu Roy, and Pranati Rakshit. 2025. Comprehensive Modernity of Fish Smoking Techniques Using the IoT., 165-190. doi:10.4018/979-83373-4332-7.ch006.

[14] R. J. Setiawan, K. Ma'ruf, D. Darmono, N. Azizah, and N. E. Khosyiati, "Systematic modernization of fish smoking method with the implementation of smoked fish machine based on Internet of Things technology," Int. J. Systematic Innovation, vol. 9, no. 1, pp. 57-67, 2025, doi: 10.6977/IJoSI.202502_9(1).0005.

APPENDIX O

CURRICULUM VITAE

Name: Denmark G. Acosta

Program: BS Computer Engineering

Mobile Nnumber: 09530457012

Email: denmarkgalacanacosta@gmail.com

PERSONAL INFORMATION
Gender: Male
Nationality: Filipino
Civil Status: Single
Religion: Born again Christian
Date of Birth: March 01, 2001
Place of Birth: Cawayan, Bongabong Oriental Mindoro
Father's Name: Dennis R. Acosta
Mother's Name: Noeme G. Acosta

Educational Background
Tertiary Level
Mindoro State University
Bachelor of Science in Computer Engineer
Labasan, Bongabong, Oriental Mindoro
Senior High School
John Paul College
Odiong, Roxas Oriental Mindoro
Junior High School
John Paul College
Odiong, Roxas Oriental Mindoro
Elementay School
Cawayan Elementary School
Cawayan, Bongabong Oriental Mindoro

Name: Jayperson M. Caya

Program: BS Computer Engineering

Address: Anilao, Bongabong, Or. Mdo.

Mobile Number: 09660296909

E-mail: jaypersonmarasigacaya@gmail.com

PERSONAL INFORMATION
Gender: Male
Nationality: Filipino
Civil Status: Single
Religion: Iglesia ni Cristo
Date of Birth: August 09,2002
Place of Birth: Calapan City, Oriental Mindoro
Father's Name: Jayson S. Caya
Mother's Name: Gloria M. Caya

EDUCATIONAL BACKGROUND

Tertiary Level
Mindoro State University
Bachelor of Science in Computer Engineering
Labasan, Bongabong, Oriental Mindoro
Senior High School
Eastern Mindor College
BB2, Bongabong, Oriental Mindoro
Junior High School
Masaguisi National High School
Masaguisi, Bongabong, Oriental Mindoro
Elementary School
Masaguisi Elementary School
Masaguisi, Bongabong, Oriental Mindoro

Name: Yasme O. Fajarito

Program: BS Computer Engineering

Address: Buhay na Tubig, Pola Or. Mdo.

Mobile Number: 09278070832

E-mail Address: fajaritoyasme@gmail.com

PERSONAL INFORMATION
Gender: Female
Nationality: Filipino
Civil Status: Single
Religion: Iglesia ni Cristo
Date of Birth: April 07, 2004
Place of Birth: Pola, Oriental Mindoro
Father's Name: Zaldy F. Fajarito
Mother's Name: Daria O. Fajarito

EDUCATIONAL BACKGROUND
Tertiary Level
Mindoro State University
Bachelor of Science in Computer Engineering
Labasan, Bongabong, Oriental Mindoro
Senior High School
Araceli B. Pantilanan, Bacawan High School
Bacawan, Pola Oriental Mindoro
Junior High School
Araceli B. Pantilanan, Bacawan High School
Bacawan, Pola Oriental Mindoro
Elementary School
Buhay na Tubig Elementary School
Buhay na Tubig, Pola Oriental Mindoro

Name: John Aeron P. Fruelda

Program: BS Computer Engineering

Address: Roxas, Oriental Mindoro

Mobile Number: 0965 774 6920

E-mail Address: JohnaeronFruelda@gmail.com

PERSONAL INFORMATION

Gender: male
Nationality: Filipino
Civil Status: Single
Religion: Born Again Christian
Date of Birth: January 07,2004
Place of Birth: Calapan City, Oriental Mindoro
Father's Name: Ruel P. Fruelda
Mother's Name: Shirley P. Fruelda

EDUCATIONAL BACKGROUND

Tertiary Level
Mindoro State University
Bachelor of Science in Computer Engineering
Labasan, Bongabong, Oriental Mindoro

Senior High School
Dangay National High School
Dangay, Roxas, Oriental Mindoro

Junior High School
Dangay National High School
Dangay, Roxas, Oriental Mindoro

Elementary School
Parian Elementary School
Parian, Calamba, Laguna

Name: Ruzzel A. Livelo

Program: BS Computer Engineering

Mobile Number: 09653469274

Emai Address: ruzzellivelo911@gmail.com

PERSONAL INFORMATION
Gender: Male
Nationality: Filipino
Civil Status: Single
Religion: Roman Catholic
Date of Birth: July 6, 2004
Place of Birth: Poblacion, Bongabong, Oriental Mindoro
Father's Name: Rommel A. Livelo
Mother's Name: Cherry Ann A. Livelo

Educational Background
Tertiary Level
Mindoro State University
Bachelor of Science in Computer Engineer
Labasan, Bongabong, Oriental Mindoro
Senior High School
Saint Joseph Academy
Bagong Bayan 2, Bongabong, Oriental Mindoro
Junior High School
Bongabong Technical and Vocational High School
Batuhan,Bongabong,Oriental Mindoro
Elementay School
Bagong Bayan 2 Elementary School
Bagong Bayan 2. Bongabong, Oriental Mindoro